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PART I

Probability Theory

1 Probability Space

1.1 Sample space, events and measurable structure

A random experiment has a *sample space* Ω of possible outcomes. An *event* is a subset of Ω, but only a prescribed family $\mathcal{F}$ of events can be assigned probabilities. A σ-*algebra* satisfies: $\Omega \in \mathcal{F}$; $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$; and $A_1, A_2, \ldots \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$. The probability space is $(\Omega, \mathcal{F}, \mathbb{P})$.

The construction emphasizes the distinction between *directly observable events* and a richer σ-algebra generated by them. For a discrete die, all subsets can be used; for an interval, the Borel σ-algebra generated by intervals is a standard event family.

1.2 Probability axioms and set limits

A probability measure satisfies $\mathbb{P}(\Omega) = 1$, nonnegativity, and countable additivity on disjoint events. Key consequences:

$$\begin{aligned} \mathbb{P}(A^c) &= 1 - \mathbb{P}(A), \\ \mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B), \\ A_n \uparrow A &\Longrightarrow \mathbb{P}(A_n) \uparrow \mathbb{P}(A), \\ A_n \downarrow A &\Longrightarrow \mathbb{P}(A_n) \downarrow \mathbb{P}(A). \end{aligned}$$

For a sequence $\{A_n\}$, the two event limits have distinct meanings:

$$\limsup_n A_n = \bigcap_{m \geq 1} \bigcup_{n \geq m} A_n \quad \text{(infinitely often)}, \qquad \liminf_n A_n = \bigcup_{m \geq 1} \bigcap_{n \geq m} A_n \quad \text{(eventually always)}.$$

Key result

Borel–Cantelli: If $\sum_n \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$. Conversely, if the A_n are independent and $\sum_n \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$.

1.3 Conditioning, total probability and Bayes' rule

For $\mathbb{P}(B) > 0$, $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$. Independence is $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, which differs from mutual exclusivity. For a partition $\{B_j\}$ with positive probabilities:

$$\mathbb{P}(A) = \sum_j \mathbb{P}(A \mid B_j)\mathbb{P}(B_j), \quad \mathbb{P}(B_k \mid A) = \frac{\mathbb{P}(A \mid B_k)\mathbb{P}(B_k)}{\sum_j \mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}.$$

Exam cue: translate a verbal observation into an event first; identify the conditioning event before invoking Bayes or independence. Poisson conditional splitting is exercised in §4 (P1–4).

1.4 The structure of an experiment: sample space, observable events and probability

An experiment has a set Ω of all possible outcomes. The description of an outcome depends on what the experiment measures. One toss of a coin has $\Omega = \{H, T\}$; one roll of a die has $\Omega = \{1, 2, 3, 4, 5, 6\}$; a

randomly chosen breaking point on a unit stick has $\Omega = [0, 1]$. A repeated experiment records an ordered tuple of outcomes. The choice of representation matters: using a relative coordinate $x/L \in [0, 1]$ for a stick of length L gives the same mathematical state space for sticks of different lengths.

Not every subset of an uncountable space need be measurable. A collection $\mathcal{F}$ of measurable events must satisfy

$$\Omega \in \mathcal{F}, \qquad A \in \mathcal{F} \Longrightarrow A^c \in \mathcal{F}, \qquad A_i \in \mathcal{F} \Longrightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}.$$

The last property, together with complements, implies closure under countable intersections. The smallest σ-algebra containing a family $\mathcal{C}$ of observable sets is $\sigma(\mathcal{C})$. For real-valued observations, intervals with rational endpoints already generate the Borel σ-algebra. Finite intersections of directly observed events allow us to build complicated events from simpler measurements.

Finite versus countable operations. An algebra is closed under *finite* unions and complements. A σ-algebra is closed under *countable* unions. Infinite-sequence events, such as "infinitely many successes occur", require the latter. The monotone limit of events can therefore be treated inside $\mathcal{F}$.

1.5 Probability measure and its calculus

A probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ is nonnegative, normalized and countably additive:

$$\mathbb{P}(\Omega) = 1, \qquad \mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i) \quad \text{for pairwise disjoint } A_i.$$

There follow monotonicity ($A \subseteq B \implies \mathbb{P}(A) \le \mathbb{P}(B)$), finite additivity, complement rules, union bounds and continuity from above/below. For every family A_i,

$$\mathbb{P}\left(\bigcup_i A_i\right) \le \sum_i \mathbb{P}(A_i), \qquad \mathbb{P}(A \cap B) \ge \mathbb{P}(A) + \mathbb{P}(B) - 1.$$

The equality $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ extends to three events through inclusion–exclusion:

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) = {} & \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ & - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).\end{aligned}$$

A continuous distribution cannot normally be specified by probabilities of singleton points: each singleton may have probability zero while the whole interval has probability one. Instead specify $\mathbb{P}((a, b])$ consistently. The cumulative distribution function is the natural extension device.

1.6 Sequences of events and Borel–Cantelli principles

For a sequence of sets, $\limsup A_n = \cap_m \cup_{n \ge m} A_n$ is the event that *infinitely many* A_n occur; $\liminf A_n = \cup_m \cap_{n \ge m} A_n$ says that *all but finitely many* occur. A set sequence converges precisely when its upper and lower limits agree. The probability continuity statements follow from countable additivity.

If $\sum_n \mathbb{P}(A_n) < \infty$, the union bound gives $\mathbb{P}(\cup_{n \ge m} A_n) \le \sum_{n \ge m} \mathbb{P}(A_n) \to 0$; letting $m \to \infty$ proves $\mathbb{P}(A_n \text{ i.o.}) = 0$. If the A_n are independent and $\sum_n \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(\cap_{n=m}^{N} A_n^c) = \prod_{n=m}^{N} [1 - \mathbb{P}(A_n)] \le \exp[-\sum_{n=m}^{N} \mathbb{P}(A_n)] \to 0$. Thus infinitely many A_n occur almost surely.

1.7 Conditional probability, chain rule and different types of independence

If $\mathbb{P}(B) > 0$, define $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$. This itself is a probability measure in A for fixed B. The multiplication rule generalizes to

$$\mathbb{P}\left(\bigcap_{j=1}^{k} A_j\right) = \mathbb{P}(A_1) \prod_{j=2}^{k} \mathbb{P}\left(A_j \mid \bigcap_{i<j} A_i\right),$$

provided the denominators are positive. For a partition $B_1, \dots, B_r$ of Ω,

$$\mathbb{P}(A) = \sum_{j=1}^{r} \mathbb{P}(A \mid B_j)\mathbb{P}(B_j), \qquad \mathbb{P}(B_i \mid A) = \frac{\mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}{\sum_j \mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}.$$

Two events are independent when $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$; pairwise independence does not imply mutual independence of three or more events. Independence and disjointness are very different: nontrivial disjoint events cannot be independent when both have positive probability.

1.8 Worked examples from finite probability models

Example 1: newspapers. Suppose $\mathbb{P}(A) = .45$, $\mathbb{P}(B) = .35$, $\mathbb{P}(C) = .30$, the pairwise intersections have probabilities $.10, .08, .05$, and the triple intersection has probability $.03$. Then

$$\mathbb{P}(A \cup B \cup C) = .45 + .35 + .30 - .10 - .08 - .05 + .03 = .90.$$

Thus 10% subscribe to none. Exactly one paper has probability

$$(.45 - .10 - .08 + .03) + (.35 - .10 - .05 + .03) + (.30 - .08 - .05 + .03) = .73.$$

Exactly two newspapers: $.10 + .08 + .05 - 3(.03) = .14$; all three: $.03$. Check $.73 + .14 + .03 = .90$.

Example 2: without-replacement sampling. A bag has five white and six black balls. In three draws without replacement,

$$\mathbb{P}(B, W, B) = \frac{6}{11} \cdot \frac{5}{10} \cdot \frac{5}{9} = \frac{5}{33}.$$

The multiplication formula automatically accounts for the changing composition.

Example 3: two dice. The state space consists of 36 ordered pairs (i, j). If $S = i + j$ is observed, the information set $\sigma(S)$ groups outcomes according to their sums. For example, conditional on $S = 7$, the six possible pairs are equally likely. Therefore $\mathbb{P}(i = 1 \mid S = 7) = 1/6$. *Conditioning changes the effective sample space.*

Example 4: sampling individuals. An outcome can be a student identification number ω. Observations such as height $H(\omega)$, weight $W(\omega)$ and home city $C(\omega)$ are separate measurement maps on the same population. Events such as "height less than 165 cm" are inverse images of intervals under H.

1.9 Pseudo-random numbers and practical simulation

Computers often create pseudo-random sequences by deterministic recursions, such as the linear congruential generator

$$Z_{n+1} = (aZ_n + c) \bmod M, \qquad U_n = \frac{Z_n}{M}.$$

The multiplier a, increment c, modulus M and seed Z_0 determine the sequence. Such numbers approximate uniform random draws for computational purposes, but a deterministic recursion is not literally an independent random experiment. A poor choice of constants produces short cycles and visible regularity. Statistical simulation evaluates the discrepancy between pseudo-random and ideal probabilistic behavior.

Example 5: a combinatorial identity by sampling. Sampling without replacement admits two descriptions: choose a subset of size a directly, or reveal objects sequentially until a stopping condition. Equating the probabilities computed by the two descriptions yields identities involving products of decreasing integers and binomial coefficients. This probabilistic technique can simplify otherwise awkward algebraic sums.

> **Key result**
>
> **Chapter 1 structure:** experiment $\to$ outcomes $\Omega \to$ observable events $\mathcal{C} \to \sigma(\mathcal{C}) \to$ probability $\mathbb{P} \to$ conditional probabilities and derived events. The fundamental object is the triple $(\Omega, \mathcal{F}, \mathbb{P})$.

1.10 Borel events, generated information and measure completion

For real-valued data, the Borel σ-algebra is generated by open intervals, closed intervals, or intervals whose endpoints are rational. These generator families lead to the same Borel events. A distribution function F specifies probabilities on half-open intervals through $\mathbb{P}((a,b]) = F(b) - F(a)$ and, with the usual right-continuity conditions, determines a unique probability measure on Borel sets.

The *completion* of a probability measure adds all subsets of events of probability zero to the measurable event family. Every Borel set is already Borel-measurable; what may be absent from the Borel σ-algebra are some subsets of null Borel sets. When the background probability is Lebesgue measure, its completion gives the Lebesgue-measurable sets. This distinction helps explain why the *event field* depends on the mathematical measurement model, rather than being the family of absolutely all subsets of the real line.

1.11 The probability derivation of a finite product identity

Suppose a population contains A balls in total, a of which are white. Draw balls without replacement until the first white ball appears. The stopping time T can take the values $1, 2, \ldots, A-a+1$. If black means unsuccessful draw, then

$$\mathbb{P}(T=k) = \prod_{j=0}^{k-2} \frac{A-a-j}{A-j} \cdot \frac{a}{A-k+1}, \qquad 1 \le k \le A-a+1,$$

where an empty product equals one. The events $\{T=k\}$ partition the sample space, so

$$\boxed{\sum_{k=1}^{A-a+1} \left(\prod_{j=0}^{k-2} \frac{A-a-j}{A-j} \right) \frac{a}{A-k+1} = 1.}$$

This identity has a direct combinatorial meaning: a first white ball must eventually be drawn, and its arrival occurs at one and only one of the listed trial numbers. It illustrates how the law of total probability can verify a nontrivial finite algebraic sum.

2 Random Variables and Distributions

2.1 Random variables are measurable maps

A random variable is a measurable function $X : \Omega \to \mathbb{R}$. The data point $x = X(\omega)$ is the realized function value. Measurability means $\{\omega : X(\omega) \le x\} \in \mathcal{F}$ for every x. The chapter's student-ID example illustrates that height, weight and other measurements are different random variables on *one* underlying probability space.

The distribution of X is characterized by the cumulative distribution function

$$F_X(x) = \mathbb{P}(X \le x), \qquad \mathbb{P}(a < X \le b) = F_X(b) - F_X(a).$$

Every CDF is nondecreasing and right-continuous, with limiting values 0 and 1 as $x \to -\infty$ and $x \to \infty$.

2.2 Discrete, continuous and singular components

For a discrete X, the probability mass $p_X(x) = \mathbb{P}(X = x) = F_X(x) - F_X(x^-)$ is a CDF jump. For an absolutely continuous X, $F_X(x) = \int_{-\infty}^{x} f_X(u)\,du$, $f_X \ge 0$, and $\int f_X = 1$. Another important case is a *singular continuous* part, exemplified by the Cantor distribution.

Cantor function. Begin with base-3 digits in $\{0, 2\}$ on the Cantor set; translate $0 \mapsto 0$, $2 \mapsto 1$ into binary digits, and extend constantly over removed middle-third intervals. The resulting CDF is continuous, non-decreasing, and has derivative zero almost everywhere, yet increases from 0 to 1. Therefore *continuous CDF* does not by itself mean *has a PDF*.

2.3 Moments, transforms and changes of variable

For integrable g, $\mathbb{E}[g(X)] = \sum_x g(x)p_X(x)$ (discrete) or $\int g(x)f_X(x)\,dx$ (density case). Define $\mu = \mathbb{E}[X]$, $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mu^2$. When finite near 0, the MGF $M_X(t) = \mathbb{E}[e^{tX}]$ satisfies $M_X'(0) = \mathbb{E}[X]$ and $M_X''(0) = \mathbb{E}[X^2]$.

If $Y = g(X)$ and g is one-to-one and differentiable, then

$$f_Y(y) = f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right|.$$

For a non-one-to-one map, sum the contributions over its inverse branches. The exercise $Y = -\log X$ for $f_X(x) = \theta x^{\theta-1}$ appears with its estimation subparts in P2–2 (§6).

Key result

Do not confuse $\mathbb{P}$ (probability of events in Ω), F_X (distribution of measured values), p_X (point mass) and f_X (density). An interval-valued random variable may still have a singular distribution.

2.4 Pushforward probabilities: from outcomes to data

An outcome $\omega \in \Omega$ may contain more information than a measurement requires. A random variable $X(\omega)$ extracts one numerical feature. For X to define measurable data, $X^{-1}((-\infty, x]) \in \mathcal{F}$ for every x. Its induced probability measure is the *pushforward*:

$$\mathbb{P}_X(B) = \mathbb{P}\{\omega : X(\omega) \in B\} = \mathbb{P}(X^{-1}(B)).$$

A random variable need not have a continuous range: a two-valued indicator on $[0,1]$ is still discrete. Conversely, a function with an interval range may induce a singular continuous law, not an ordinary density.

2.5 Distribution functions: atomic, absolutely continuous and singular laws

A CDF satisfies monotonicity, right continuity and $F(-\infty) = 0, F(+\infty) = 1$. At a point a, its jump is $F(a) - F(a^-) = \mathbb{P}(X = a)$. If $F(x) = \int_{-\infty}^{x} f(t)dt$ for some nonnegative integrable f, the probability of X lying in $(a, b]$ is $\int_a^b f(t)dt$, and singletons have zero probability. The CDF may be decomposed into atomic, absolutely continuous and singular continuous parts.

The *Cantor distribution* illustrates the last case. Repeatedly remove the middle third of each surviving interval. The Cantor set has length zero since the total removed length is $\sum_{k=0}^{\infty} 2^k/3^{k+1} = 1$. Assigning binary digits to surviving ternary digits gives the devil's-staircase CDF: it is nondecreasing and continuous, goes from 0 to 1, and is locally constant on all removed intervals. Its derivative vanishes Lebesgue-almost everywhere, yet it is not constant. Thus $\int_0^1 F'(x)dx = 0 \neq F(1) - F(0) = 1$.

2.6 Distribution construction and inverse transforms

If $U \sim \mathrm{Uniform}(0,1)$ and F is a CDF, define the generalized inverse

$$F^{-1}(u) = \inf\{x : F(x) \geq u\}.$$

Then $X = F^{-1}(U)$ has CDF F, since $\{F^{-1}(U) \leq x\} = \{U \leq F(x)\}$ up to null events. This is the *inverse-CDF method* of simulation. It works for discrete and continuous laws without assuming a differentiable inverse.

For a differentiable strictly monotone map $Y = g(X)$, use F_Y or the change-of-variables formula

$$f_Y(y) = f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right|.$$

For a non-injective map, add contributions from all valid inverse branches. For example, $Y = X^2$ with X continuous gives

$$f_Y(y) = \frac{f_X(\sqrt{y}) + f_X(-\sqrt{y})}{2\sqrt{y}}, \qquad y > 0.$$

When the parameter affects support, keep support indicators while deriving a likelihood: discarding them can change the maximizing parameter.

2.7 Expectation, moments and central tendency

For a real-valued X, $\mathbb{E}[g(X)]$ means integration against its distribution, $\int_{\mathbb{R}} g(x)dF_X(x)$. When the positive and negative parts are both infinite, the expectation does not exist, even if a symmetric principal-value integral equals zero.

$$\mathbb{E}[X^k] = \int x^k dF_X(x), \quad \mu = \mathbb{E}X, \quad \operatorname{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mu^2.$$

Linear expectation does not require independence: $\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y$. By contrast, $\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y)$, so independence becomes relevant when simplifying variances.

Cauchy example. The standard Cauchy density is $f(x) = 1/[\pi(1+x^2)]$. Its formal integral $\int xf(x)dx$ is not absolutely convergent; the positive and negative parts are both infinite. There is no finite $\mathbb{E}X$, and symmetry is not enough to claim $\mathbb{E}X = 0$. In fact even its first absolute moment is infinite.

2.8 Moment generating, characteristic and cumulant functions

Define $M_X(t) = \mathbb{E}e^{tX}$ where finite, the characteristic function $\varphi_X(t) = \mathbb{E}e^{itX}$ (always defined), and the cumulant $K_X(t) = \log M_X(t)$ where M_X exists. Differentiation at zero gives

$$M_X^{(k)}(0) = \mathbb{E}X^k, \qquad K_X'(0) = \mathbb{E}X, \qquad K_X''(0) = \operatorname{Var}(X).$$

For independent variables X, Y, $M_{X+Y}(t) = M_X(t)M_Y(t)$ and $K_{X+Y} = K_X + K_Y$. This is why moment-generating functions are convenient for proving sum laws and the central limit theorem. Two distributions with MGFs equal on a neighborhood of zero have the same law; characteristic functions are more general and also determine distributions uniquely.

2.9 Standard univariate distribution reference

Law	Mass or density on its support	Mean / variance
Bernoulli(p)	$p^x(1-p)^{1-x}$, $x \in \{0,1\}$	p; $p(1-p)$
Binomial(n,p)	$\binom{n}{k}p^k(1-p)^{n-k}$, $0 \le k \le n$	np; $np(1-p)$
Geometric(p)	$p(1-p)^{k-1}$, $k \ge 1$	$1/p$; $(1-p)/p^2$
Poisson(λ)	$e^{-\lambda}\lambda^k/k!$, $k \ge 0$	λ; λ
Uniform(a,b)	$(b-a)^{-1}$, $a < x < b$	$(a+b)/2$; $(b-a)^2/12$
Exponential rate λ	$\lambda e^{-\lambda x}$, $x \ge 0$	$1/\lambda$; $1/\lambda^2$
Gamma shape k, rate λ	$\lambda^k x^{k-1}e^{-\lambda x}/\Gamma(k)$, $x > 0$	k/λ; k/λ^2
Normal(μ,σ^2)	$(\sigma\sqrt{2\pi})^{-1}e^{-(x-\mu)^2/(2\sigma^2)}$	μ; σ^2

Key result

Recognizing the model: fixed trials and success count ⇒ binomial; waiting for a success ⇒ geometric; rare counts in time ⇒ Poisson; waiting time of an event ⇒ exponential; waiting for the kth event ⇒ gamma. The normal law describes standardized sums and many limit distributions.

2.10 From a sample to a histogram and an estimated density

For observed data $x_1, \dots, x_n$, let the order statistics be $x_{(1)} \le \cdots \le x_{(n)}$. A histogram divides the observed range into bins of width h, counts the number n_j of observations in bin j, and plots bar height $n_j/(nh)$. The height is divided by bin width so that total area is one:

$$\widehat{f}_h(x) = \frac{n_j}{nh} \quad \text{inside bin } j, \qquad \int \widehat{f}_h(x)dx = \sum_j \frac{n_j}{n} = 1.$$

For a small interval $(a, a+h]$, the expected relative frequency is $F(a+h) - F(a)$, so the expected density height is approximately $[F(a+h) - F(a)]/h \approx f(a)$ when a density exists and h is small. Larger bins smooth random variation but erase detail; narrower bins preserve detail but have greater sampling noise. This is a concrete instance of the bias–variance tradeoff in distribution estimation.

2.11 Descriptive sample statistics as partial information

The entire distribution conveys more information than any fixed finite list of moments. In practice, a few sample statistics summarize key features of observed data. The sample mean and median measure location; the variance and standard deviation measure dispersion; skewness measures asymmetry; kurtosis measures tail or outlier sensitivity.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2},$$

$$m'_k = \frac{1}{n}\sum_{i=1}^{n} x_i^k, \qquad \widehat{\gamma}_1 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^3, \qquad \widehat{\gamma}_2 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^4 - 3.$$

The -3 in excess kurtosis makes the population normal distribution's excess kurtosis zero. Definitions using n rather than $n-1$ in moment denominators may vary; when solving a particular exercise use the convention explicitly given.

The sample median is the middle order statistic when n is odd, and the average of the two middle order statistics when n is even:

$$\widetilde{x} = \begin{cases} x_{((n+1)/2)}, & n \text{ odd}, \\ \frac{1}{2}[x_{(n/2)} + x_{(n/2+1)}], & n \text{ even}. \end{cases}$$

Unlike the sample mean, the median is insensitive to the magnitude of a small number of extreme observations. Under moment-existence assumptions, the LLN also justifies convergence of empirical raw moments $n^{-1}\sum X_i^k$ to $\mathbb{E}[X^k]$.

2.12 Worked distribution examples on the unit probability space

Take $U \sim \text{Uniform}(0,1)$. Different measurable maps of this *same* uniform variable can have completely different distributions.

Indicator map. If $X = \mathbf{1}\{U \le p\}$ for $0 < p < 1$, then $\mathbb{P}(X=1) = p$, $\mathbb{P}(X=0) = 1-p$, and

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1-p, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

Thus the discontinuities of the CDF are exactly its positive-mass points.

Sine map and the arcsine law. Let $X = \sin(\pi U)$. For $0 < x < 1$, the event $X \le x$ is the union of two end intervals of total length $2\arcsin(x)/\pi$, giving

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{2}{\pi}\arcsin x, & 0 \le x \le 1, \\ 1, & x > 1, \end{cases} \qquad f_X(x) = \frac{2}{\pi\sqrt{1-x^2}}, \quad 0 < x < 1.$$

The two inverse branches explain the factor of two; a one-branch Jacobian would lose half the probability.

Arbitrary target distribution. If F is any CDF, applying the generalized inverse F^{-1} to uniform U produces a variable with CDF F. The indicator, sine and inverse-quantile examples show why the range of a measurement map alone does not determine whether the induced law is discrete, absolutely continuous or singular.

2.13 When moment methods describe and when they fail

In theory, the CDF describes a distribution completely, while low-order moments describe only partial information. Some distributions are uniquely determined by their entire moment sequences under suitable conditions, but equal means and variances alone never guarantee identical laws. Some laws have no finite moments at all, as with the standard Cauchy distribution. In estimation, a moment estimator can be convenient even when it is less efficient than an MLE; a moment-matching formula must be checked against existence and identifiability of the population moments.

3 Coin Tossing and Related Distributions

3.1 An infinite Bernoulli process

Let $\{B_i\}_{i\geq 1}$ be i.i.d. Bernoulli(p): $\mathbb{P}(B_i = 1) = p$ and $\mathbb{P}(B_i = 0) = 1 - p$. An infinite outcome is a binary sequence $\omega = (\omega_1, \omega_2, \ldots)$. Finite-coordinate (cylinder) events generate its event σ-algebra.

For $B \sim \text{Bern}(p)$, $\mathbb{E}[B] = p$, $\text{Var}(B) = p(1-p)$ and $M_B(t) = 1 - p + pe^t$.

3.2 Counting successes and waiting times

Quantity	Distribution / PMF	Mean; variance
$N_n = \sum_{i=1}^{n} B_i$	Bin(n,p): $\binom{n}{k}p^k(1-p)^{n-k}$, $k = 0, \ldots, n$	np; $np(1-p)$
W_1: first success trial	$\mathbb{P}(W_1 = k) = (1-p)^{k-1}p$, $k \geq 1$	$1/p$; $(1-p)/p^2$
W_r: rth success trial	$\mathbb{P}(W_r = k) = \binom{k-1}{r-1}p^r(1-p)^{k-r}$, $k \geq r$	r/p; $r(1-p)/p^2$

The *counting process* $N(t)$ and *waiting times* W_k are dual: $\{N(t) \geq k\} = \{W_k \leq t\}$. The multinomial extension is treated in §4.

3.3 Rare events and continuous time

For $N_n \sim \text{Bin}(n, p_n)$, if $p_n \to 0$ and $np_n \to \lambda$, then $N_n \xrightarrow{d}$ Poisson(λ) with $\mathbb{P}(N = k) = e^{-\lambda}\lambda^k/k!$. A properly rescaled geometric waiting time converges to an exponential waiting time. Sums of r independent exponential waiting times form a gamma distribution. In a Poisson process with intensity λ, $N(t) \sim \text{Poisson}(\lambda t)$ and $W_k \sim \text{Gamma}(k, \text{rate } \lambda)$.

> **Key result**
>
> **Useful bridge:** Poisson counts $\leftrightarrow$ gamma waiting times $\leftrightarrow$ chi-square variables ($2\lambda W_k \sim \chi^2_{2k}$). This bridge is the entire method of the exact Poisson interval F–8 in §7.

3.4 Normal approximation

For fixed $0 < p < 1$, the standardized binomial obeys

$$\frac{N_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} N(0,1).$$

More general LLN and CLT statements, including MGF proof exercises, appear in §5. A related topic is Gaussian density via polar coordinates and logarithmic / Chernoff-type probability bounds.

3.5 Finite-cylinder construction of an infinite coin sequence

An infinite tossing process is modeled by $\Omega = \{0,1\}^{\mathbb{N}}$. A cylinder $C_{i_1\cdots i_n}$ fixes the first n toss outcomes and leaves future outcomes unrestricted. If a success has probability p, then

$$\mathbb{P}(C_{i_1\cdots i_n}) = p^{\sum_{j=1}^n i_j}(1-p)^{n-\sum_{j=1}^n i_j}.$$

The consistency identity $\mathbb{P}(C_{i_1\cdots i_n}) = \mathbb{P}(C_{i_1\cdots i_n 0}) + \mathbb{P}(C_{i_1\cdots i_n 1})$ permits the probabilities to extend to the event σ-algebra generated by finite observations. The coordinate maps $B_k(\omega) = \omega_k$ are i.i.d. Bernoulli random variables.

3.6 Two descriptions of the same process: counts and hitting times

The number of successes among the first n tosses is $N_n = \sum_{i=1}^n B_i$. For $0 \le k \le n$, the number of sequences with exactly k successes is $\binom{n}{k}$; therefore

$$\mathbb{P}(N_n = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad M_{N_n}(t) = (1-p+pe^t)^n.$$

Different disjoint blocks of tosses yield independent binomial counts. Writing W_k for the trial number of the kth success,

$$\{W_k \le n\} = \{N_n \ge k\}, \qquad \mathbb{P}(W_k = n) = \binom{n-1}{k-1} p^k (1-p)^{n-k}, \quad n \ge k.$$

The first-success time is geometric with parameter p. It is memoryless: $\mathbb{P}(W_1 > m+n \mid W_1 > m) = \mathbb{P}(W_1 > n) = (1-p)^n$. Independent increments $W_k - W_{k-1}$ are geometric, so

$$\mathbb{E}W_k = \frac{k}{p}, \qquad \mathrm{Var}(W_k) = \frac{k(1-p)}{p^2}, \qquad M_{W_k}(t) = \left(\frac{pe^t}{1-(1-p)e^t}\right)^k$$

for $t < -\log(1-p)$.

3.7 Rare-event scaling: geometric to exponential, binomial to Poisson

Let a trial occupy time step Δ and set $p = \lambda\Delta$ with $\Delta \downarrow 0$. The geometric waiting time in physical units is $T_\Delta = \Delta W_1$. Its survival probability satisfies

$$\mathbb{P}(T_\Delta > t) = (1-\lambda\Delta)^{\lfloor t/\Delta \rfloor} \longrightarrow e^{-\lambda t}.$$

Hence $T_\Delta \Rightarrow \mathrm{Exp}(\text{rate } \lambda)$. Similarly, when $n \to \infty$, $p_n \to 0$ and $np_n \to \lambda$, the binomial probability satisfies

$$\binom{n}{k} p_n^k (1-p_n)^{n-k} \longrightarrow \frac{e^{-\lambda}\lambda^k}{k!},$$

for every fixed k. The limiting count is Poisson(λ). Its MGF is $\exp\{\lambda(e^t-1)\}$.

3.8 Continuous-time Poisson process, gamma waiting time and counting measure

A homogeneous Poisson process $\{N(t) : t \ge 0\}$ with rate λ starts at $N(0) = 0$, has independent increments, stationary increments, and

$$N(t) - N(s) \sim \mathrm{Poisson}(\lambda(t-s)), \quad 0 \le s < t.$$

The inter-arrival times E_i are independent exponential variables with rate λ. The time of the kth arrival $G_k = \sum_{j=1}^{k} E_j$ has density

$$f_{G_k}(t) = \frac{\lambda^k t^{k-1} e^{-\lambda t}}{\Gamma(k)}, \quad t > 0.$$

For integer k, $\Gamma(k) = (k-1)!$. The relationship $\{N(t) \geq k\} = \{G_k \leq t\}$ is central to Poisson tail identities and confidence intervals. More generally, the number of events falling in a set A is a *counting measure*, $N(A) = \sum_j \mathbf{1}\{G_j \in A\}$. Disjoint intervals have independent Poisson counts, and $\mathbb{E}N((s,t]) = \lambda(t-s)$.

Example: customer arrivals. Suppose the expected number of customers is 30 per hour. The event that fewer than two customers arrive in the first 5 minutes is the event $N(1/12) \leq 1$, with mean $30/12 = 2.5$. Thus the probability that the second customer has not arrived after five minutes is

$$\mathbb{P}(G_2 > 1/12) = \mathbb{P}(N(1/12) \leq 1) = e^{-2.5}(1 + 2.5).$$

Example: collecting four cereal prizes. Each box contains one of four equally likely prizes. The first distinct prize arrives in one box. After one distinct prize, the chance of a new one is $3/4$; after two, it is $2/4$; after three, $1/4$. If G_j denote the successive geometric waiting times, linearity yields

$$\mathbb{E}\left[\sum_{j=1}^{4} G_j\right] = 1 + \frac{4}{3} + \frac{4}{2} + 4 = \frac{25}{3}.$$

This is the simplest coupon-collector example.

3.9 Gamma, beta and chi-square relations

If $U \sim \Gamma(a, \text{scale } b)$ and $V \sim \Gamma(c, \text{scale } b)$ are independent, then $U + V \sim \Gamma(a + c, \text{scale } b)$ and $U/(U+V) \sim \text{Beta}(a, c)$. The ratio is independent of the sum. A standard normal squared is $Z^2 \sim \chi^2_1 = \Gamma(1/2, \text{scale } 2)$; $\sum_{j=1}^{k} Z_j^2 \sim \chi^2_k$. In particular,

$$2\lambda G_k \sim \chi^2_{2k}.$$

This gamma/chi-square bridge is useful whenever an exact pivot must be constructed.

3.10 Long-time counts, scaling and the normal law

The LLN and CLT for Bernoulli trials give

$$\frac{N_n}{n} \xrightarrow{a.s.} p, \qquad \frac{N_n - np}{\sqrt{np(1-p)}} \Rightarrow N(0,1).$$

The unscaled count diverges with n when $p > 0$. Centering by np and scaling by the standard deviation reveals a non-degenerate limiting law. For independent Poisson counts $N(t)$ with $t \to \infty$, the analogous statement is

$$\frac{N(t) - \lambda t}{\sqrt{\lambda t}} \Rightarrow N(0,1).$$

A binomial probability for $N_n/n \geq x$ can be studied by the binary relative entropy

$$D(x\|p) = x \log \frac{x}{p} + (1-x) \log \frac{1-x}{1-p}.$$

It satisfies $D(p\|p) = D'(p\|p) = 0$ and $D''(p\|p) = 1/[p(1-p)]$, so locally $D(x\|p) \approx (x-p)^2/[2p(1-p)]$. The quadratic expansion explains Gaussian fluctuations; its full expression gives sharper exponential tail bounds.

3.11 Normal integral and distributions obtained by transformation

The standard normal density is $\phi(x) = (2\pi)^{-1/2} e^{-x^2/2}$. Its normalization follows from the two-dimensional Gaussian integral. If $I = \int_{\mathbb{R}} e^{-x^2/2} dx$, then polar coordinates yield

$$I^2 = \int_{\mathbb{R}^2} e^{-(x^2+y^2)/2}\, dx\, dy = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r\, dr\, d\vartheta = 2\pi,$$

so $I = \sqrt{2\pi}$. The family $N(\mu, \sigma^2)$ follows by the affine transformation $X = \mu + \sigma Z$.

Lognormal example. If $Y \sim N(0,1)$ and $X = e^Y$, then $X > 0$ and

$$F_X(x) = \Phi(\log x), \quad f_X(x) = \frac{1}{x\sqrt{2\pi}} e^{-(\log x)^2/2}, \quad \mathbb{E}[X^r] = \mathbb{E}e^{rY} = e^{r^2/2}.$$

The same transformation technique handles nonzero normal mean and variance by replacing $\log x$ with $(\log x - \mu)/\sigma$.

> **Key result**
>
> **Process dictionary:** Bernoulli counts are binomial; their discrete waiting times are geometric/negative binomial. Under rare-event time scaling, the limiting counts are Poisson and waiting times are exponential/gamma. Under long-time centering and standardization, counts are approximately normal.

3.12 Simulation examples: empirical process shapes

The counting and waiting-time laws can be verified computationally, not because simulation proves a theorem but because large independent samples reveal its predictions.

Simulating exponential waiting time. Generate many geometric variables $W_j \sim \text{Geom}(p)$ for small p, and plot the histogram of the rescaled times pW_j with density normalization. Since $\mathbb{P}(pW_1 > t) = (1-p)^{\lfloor t/p \rfloor} \to e^{-t}$, the histogram approaches the density $e^{-t}\mathbf{1}\{t > 0\}$ as $p \to 0$. Using the same raw geometric outcomes without rescaling would shift the entire histogram to larger trial numbers instead of producing a stable limit.

Simulating a Poisson count. For an event rate λ, choose success probability $p = \lambda/n$ for a large number of Bernoulli trials. The count of successes has $\text{Bin}(n, \lambda/n)$ law, close to $\text{Poi}(\lambda)$. Repeating the simulation for successively larger n with np fixed demonstrates the rare-event limit. Plotting the count process over increasing time also reveals approximately independent increments.

Simulating the normal approximation. With fixed $p = 0.1$ and increasing sample sizes such as $n = 300, 3000, 30000$, simulate $\text{Bin}(n, p)$ counts and center/scale by np and $\sqrt{np(1-p)}$. The standardized histograms increasingly resemble a standard normal bell curve. Histograms of unscaled counts do not overlay because the mean and variance both grow with n.

Reproducibility. In pseudo-random computer experiments, setting a fixed seed makes the generated numerical output repeatable. A seed controls the deterministic sequence used by a pseudo-random generator; it does not change the mathematical distribution stipulated for the ideal random model.

4 Multivariate Random Variables and Conditional Information

4.1 Joint distributions, covariance and transforms

For a vector $\boldsymbol{X} = (X_1, \ldots, X_d)$, the joint CDF is $F(\boldsymbol{x}) = \mathbb{P}(X_1 \le x_1, \ldots, X_d \le x_d)$. A joint density integrates to marginal densities; independence means $F(\boldsymbol{x}) = \prod_j F_j(x_j)$ (equivalently a factorized joint density when densities exist).

The covariance matrix $\Sigma = \mathbb{E}[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top]$ is symmetric positive semidefinite. For constants a, b,

$\operatorname{Var}(aX+bY) = a^2 \operatorname{Var} X + b^2 \operatorname{Var} Y + 2ab \operatorname{Cov}(X,Y)$. The multivariate MGF $M_{\boldsymbol{X}}(\boldsymbol{t}) = \mathbb{E}[e^{\boldsymbol{t}^\top \boldsymbol{X}}]$ and cumulant $K = \log M$ encode moments; $\nabla K(0) = \boldsymbol{\mu}$ and $\nabla^2 K(0) = \Sigma$.

4.2 Multinomial distribution and transformation

If a repeated trial takes categories $1, \dots, r$ with probabilities p_j and $\sum_j p_j = 1$, the count vector $(N_1, \dots, N_r)$ is multinomial:

$$\mathbb{P}(N_j = n_j,\ j = 1, \dots, r) = \frac{n!}{\prod_j n_j!} \prod_j p_j^{n_j}, \quad \sum_j n_j = n.$$

$\mathbb{E}[N_i] = np_i$, $\operatorname{Var}(N_i) = np_i(1-p_i)$, and $\operatorname{Cov}(N_i, N_j) = -np_ip_j$ for $i \neq j$. For an invertible differentiable transformation $\boldsymbol{Y} = h(\boldsymbol{X})$,

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = f_{\boldsymbol{X}}(h^{-1}(\boldsymbol{y}))\ \left|\det Dh^{-1}(\boldsymbol{y})\right|.$$

4.3 Conditional expectation is information projection

The σ-algebra perspective defines $\mathbb{E}[X \mid \mathcal{G}]$ as a $\mathcal{G}$-measurable random variable with $\int_A \mathbb{E}[X \mid \mathcal{G}] d\mathbb{P} = \int_A X d\mathbb{P}$ for every $A \in \mathcal{G}$. Key rules include $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$ and, for $\mathcal{G}_1 \subseteq \mathcal{G}_2$, the tower property

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}_2] \mid \mathcal{G}_1] = \mathbb{E}[X \mid \mathcal{G}_1].$$

Conditioning helps eliminate nuisance parameters and is central to Rao–Blackwell improvements (P2–5).

4.4 Multivariate measurements: joint, marginal and conditional distributions

A random vector is a measurable map $\boldsymbol{X} = (X_1, \dots, X_d) : \Omega \to \mathbb{R}^d$. Its joint CDF is

$$F_{\boldsymbol{X}}(x_1, \dots, x_d) = \mathbb{P}(X_1 \le x_1, \dots, X_d \le x_d).$$

If there is a density $f(x_1, \dots, x_d)$, integration over the remaining coordinates gives a marginal density, for example $f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy$. When $f_X(x) > 0$, the conditional density is $f_{Y|X}(y \mid x) = f(x,y)/f_X(x)$. These are different objects: a joint law describes all measurements simultaneously, a marginal law describes one feature, and a conditional law describes what remains uncertain after another feature is known.

Random variables X and Y are independent if for any Borel sets A, B,

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B).$$

For densities, this is equivalent to $f_{X,Y}(x,y) = f_X(x) f_Y(y)$ almost everywhere. Equal or zero covariance alone is not enough to imply independence outside special model classes.

4.5 Covariance geometry, Cauchy–Schwarz and Jensen

The covariance $\operatorname{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$ satisfies

$$|\operatorname{Cov}(X,Y)| \le \sqrt{\operatorname{Var}(X) \operatorname{Var}(Y)}.$$

To see the Cauchy–Schwarz inequality, note that $\mathbb{E}[(sX+Y)^2] \ge 0$ for every real s. The discriminant of this quadratic in s must be nonpositive, yielding $(\mathbb{E}XY)^2 \le \mathbb{E}X^2 \mathbb{E}Y^2$. For a vector $\boldsymbol{X}$ the covariance matrix Σ is positive semidefinite because for every vector a, $a^\top \Sigma a = \operatorname{Var}(a^\top \boldsymbol{X}) \ge 0$.

Jensen's inequality says $h(\mathbb{E}X) \le \mathbb{E}h(X)$ for convex h; the direction reverses for concave h. For instance, $h(x) = \sqrt{x}$ on nonnegative values gives $\mathbb{E}\sqrt{X} \le \sqrt{\mathbb{E}X}$. This is the main one-line argument in the normal sample standard-deviation exercise.

4.6 Multivariate MGFs and multinomial counts

For $\boldsymbol{X} \in \mathbb{R}^d$, $M_{\boldsymbol{X}}(t) = \mathbb{E}[e^{t^\top \boldsymbol{X}}]$ and $K(t) = \log M(t)$. Provided the derivatives exist,

$$\left.\frac{\partial K}{\partial t_j}\right|_0 = \mathbb{E}X_j, \qquad \left.\frac{\partial^2 K}{\partial t_i \partial t_j}\right|_0 = \operatorname{Cov}(X_i, X_j).$$

In repeated dice trials, the six category counts $N_1, \dots, N_6$ sum to n. For arbitrary category probabilities $p_1 + \cdots + p_6 = 1$ they have a multinomial law with MGF

$$M_{\boldsymbol{N}}(t_1, \dots, t_6) = \left(\sum_{j=1}^{6} p_j e^{t_j}\right)^n.$$

The category counts are not independent because their sum is fixed: for $i \neq j$, $\operatorname{Cov}(N_i, N_j) = -np_ip_j$. Their normalized empirical frequencies N_j/n estimate the category probabilities. By the multivariate CLT, the centered vector $\sqrt{n}(\boldsymbol{N}/n - \boldsymbol{p})$ converges to a multivariate normal with covariance $\operatorname{diag}(p) - pp^\top$ (singular because frequencies sum to one).

4.7 Multivariate normal law and linear maps

A multivariate normal vector $\boldsymbol{X} \sim N_d(\mu, \Sigma)$ has characteristic function $\mathbb{E}[e^{it^\top X}] = \exp(it^\top\mu - \frac{1}{2}t^\top\Sigma t)$. When Σ is positive definite, its density is

$$f_{\boldsymbol{X}}(x) = \frac{\exp[-\frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)]}{(2\pi)^{d/2}\sqrt{\det\Sigma}}.$$

For a deterministic matrix A and vector b, $A\boldsymbol{X} + b \sim N(A\mu, A\Sigma A^\top)$. The covariance matrix determines ellipse orientation, scale and dependence. For a jointly Gaussian vector, zero covariance between components implies independence; this implication fails in general.

4.8 Transformation of random vectors and Jacobians

For an invertible differentiable map $\boldsymbol{Y} = h(\boldsymbol{X})$,

$$f_{\boldsymbol{Y}}(y) = f_{\boldsymbol{X}}(h^{-1}(y))\left|\det Dh^{-1}(y)\right|.$$

The determinant measures the factor by which small volume elements are stretched. If a map has multiple inverse branches, apply the Jacobian on each branch and sum. For planar polar coordinates $X = R\cos\Theta$, $Y = R\sin\Theta$, the area element is $dx\,dy = r\,dr\,d\theta$ and the Jacobian is r. This also gives the polar-coordinate proof of the Gaussian integral.

4.9 Conditioning on events and conditioning on information

Conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is characterized by two requirements: it is $\mathcal{G}$-measurable, and $\mathbb{E}[X\mathbf{1}_A] = \mathbb{E}[\mathbb{E}(X \mid \mathcal{G})\mathbf{1}_A]$ for every $A \in \mathcal{G}$. For a finite partition $\{B_1, \dots, B_m\}$,

$$\mathbb{E}[X \mid \sigma(B_1, \dots, B_m)] = \sum_{j=1}^{m} \mathbb{E}[X \mid B_j]\mathbf{1}_{B_j}.$$

If Y is a measurement, one often writes $\mathbb{E}[X \mid Y] = g(Y)$, where $g(y) = \mathbb{E}[X \mid Y = y]$ is a version of the conditional mean. Uniqueness holds almost surely, not necessarily pointwise for every y.

The most important algebraic rules are

$$\begin{aligned}
\mathbb{E}[\mathbb{E}(X \mid \mathcal{G})] &= \mathbb{E}X, \\
\mathbb{E}[ZX \mid \mathcal{G}] &= Z\mathbb{E}[X \mid \mathcal{G}] \quad \text{if } Z \text{ is } \mathcal{G}\text{-measurable and integrable}, \\
\mathbb{E}[\mathbb{E}(X \mid \mathcal{G}_2) \mid \mathcal{G}_1] &= \mathbb{E}[X \mid \mathcal{G}_1] \quad (\mathcal{G}_1 \subseteq \mathcal{G}_2), \\
\mathbb{E}[X \mid \mathcal{G}] &= \mathbb{E}X \quad \text{if } X \text{ is independent of } \mathcal{G}.
\end{aligned}$$

For square-integrable X, conditional expectation is also the mean-square optimal predictor based on $\mathcal{G}$: $\mathbb{E}[(X-Z)^2]$ is minimized over $\mathcal{G}$-measurable Z by $Z=\mathbb{E}[X\mid\mathcal{G}]$. In the language of statistics, conditioning can remove irrelevant randomness while preserving unbiasedness.

Law of total variance. If $\mathbb{E}X^2<\infty$,

$$\operatorname{Var}(X)=\mathbb{E}[\operatorname{Var}(X\mid\mathcal{G})]+\operatorname{Var}(\mathbb{E}[X\mid\mathcal{G}]).$$

The first term is residual uncertainty after learning $\mathcal{G}$; the second is uncertainty in the conditional mean across information states.

> **Key result**
>
> **Core links:** joint distribution $\to$ marginals and conditionals; MGF $\to$ mean/covariance; Jacobian $\to$ transformed distributions; conditional expectation $\to$ tower property and optimal prediction; Poisson conditioning $\to$ a binomial law free of the common intensity.

4.10 Exercise P1–4: Poisson splitting and a conditional test

Exercise P1–4 Independent Poisson components and two-district counts

Prompt. Let independent nonnegative integer-valued X, Y have $Z = X + Y$. (a) Relate independent Poisson laws and the conditional law $X \mid Z = n \sim \text{Bin}(n, p)$, with the reverse characterization under the stated assumptions. (b) District counts are 63 and 81, modeled as independent Poissons. Test equality of means at level 0.05.

Core calculation. If $X \sim \text{Poisson}(\lambda_1)$, $Y \sim \text{Poisson}(\lambda_2)$ independent, then $Z \sim \text{Poisson}(\lambda_1 + \lambda_2)$ and

$$\mathbb{P}(X = x \mid Z = n) = \binom{n}{x} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^x \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-x}.$$

For the converse, the independence assumption and a common fixed $p \in (0, 1)$ imply a functional equation for the probability generating functions $G_X(s)G_Y(t) = G_Z(ps + (1-p)t)$, whose nondegenerate solutions are Poisson generating functions. Merely substituting a guessed exponential PGF checks the answer but is not a complete converse proof.

Test. Conditional on $Z = 144$ and $H_0 : \lambda_A = \lambda_B$, $X_A \mid Z = 144 \sim \text{Bin}(144, 1/2)$. The observed 63 is 1.5 standard deviations below 72: $(63 - 72)/6 = -1.5$. A normal two-sided approximation without continuity correction gives $p \approx 0.134$; the exact conditional two-sided binomial tail is $2\mathbb{P}(\text{Bin}(144, 1/2) \le 63) \approx 0.1563$. **Do not reject** H_0 at 5%.

Detailed converse proof. Write $a_x = \mathbb{P}(X = x)$, $b_y = \mathbb{P}(Y = y)$, $c_n = \mathbb{P}(Z = n)$, and $q = 1 - p$. The conditional binomial assumption and independence yield

$$a_x b_y = c_{x+y} \binom{x+y}{x} p^x q^y, \qquad x, y \ge 0.$$

Assume $0 < p < 1$ and the indicated probabilities are positive. Taking $y = 0$ and $x = 0$ gives $a_x b_0 = c_x p^x$, $a_0 b_y = c_y q^y$ and $a_0 b_0 = c_0$. Multiplying the first two identities and using $a_0 b_0 = c_0$ gives

$$c_{x+y} \binom{x+y}{x} = \frac{c_x c_y}{c_0}.$$

Set $d_n = n! c_n / c_0$. Then $d_{x+y} = d_x d_y$, $d_0 = 1$, so by induction $d_n = d_1^n$. Consequently $c_n = c_0 d_1^n / n!$. Summing over n gives $c_0 = e^{-d_1}$, so $Z \sim \text{Poisson}(d_1)$. The identities above then imply $X \sim \text{Poisson}(pd_1)$ and $Y \sim \text{Poisson}(qd_1)$; this proves the reverse characterization, not merely consistency with a guess.

Exact two-sided test. Under H_0, conditioning removes the unknown common rate. The exact symmetric null tail is $2\sum_{k=0}^{63} \binom{144}{k} 2^{-144} = 0.156339842\ldots$. As the value exceeds 0.05, the data do not give evidence of unequal rates at the stated level. A normal approximation with no continuity correction gives 0.1336; with continuity correction it is closer to the exact result. State which convention is used.

5 Convergence, LLN, CLT and Monte Carlo

5.1 Four convergence modes

The chapter proceeds from convergence of sample paths to convergence of distributions:

$$\begin{aligned} X_n \to X \text{ pointwise}: &\quad X_n(\omega) \to X(\omega) \text{ for every } \omega, \\ X_n \xrightarrow{a.s.} X: &\quad \mathbb{P}\{\omega : X_n(\omega) \to X(\omega)\} = 1, \\ X_n \xrightarrow{p} X: &\quad \mathbb{P}(|X_n - X| > \varepsilon) \to 0 \ \forall \varepsilon > 0, \\ X_n \xrightarrow{d} X: &\quad F_{X_n}(x) \to F_X(x) \text{ at continuity points of } F_X. \end{aligned}$$

Pointwise $\Rightarrow$ almost sure $\Rightarrow$ probability $\Rightarrow$ distribution, with reverse arrows false in general. A useful special case is: if $X_n \xrightarrow{d} c$ for a constant c, then $X_n \xrightarrow{p} c$.

5.2 The law of large numbers and central limit theorem

For i.i.d. integrable X_i with $\mathbb{E}[X_i] = \mu$, the strong law gives $\bar{X}_n \xrightarrow{a.s.} \mu$. If $0 < \mathrm{Var}(X_i) = \sigma^2 < \infty$, the CLT gives

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1).$$

The LLN answers *where the mean settles*; the CLT answers *how its error fluctuates on the* $n^{-1/2}$ *scale*. Chebyshev's bound is $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2)$.

5.3 Slutsky, continuous mapping and Monte Carlo

If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, then $X_nY_n \xrightarrow{d} cX$ and, if $c \neq 0$, $X_n/Y_n \xrightarrow{d} X/c$. For continuous g, $X_n \xrightarrow{d} X$ implies $g(X_n) \xrightarrow{d} g(X)$ (similarly in probability). Monte Carlo estimates $I = \mathbb{E}[g(U)]$ by $\hat{I}_n = n^{-1}\sum_i g(U_i)$; use LLN for consistency and CLT or concentration inequalities to quantify error. The PAC framing asks for $\mathbb{P}(|\hat{I}_n - I| > \varepsilon) \leq \delta$.

5.4 Modes of convergence in detail

A sequence of random variables can converge in different senses. *Pointwise* convergence means $X_n(\omega) \to X(\omega)$ for every ω; *almost sure* convergence permits an exceptional event of probability zero. *Convergence in probability* requires $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for each fixed $\varepsilon > 0$. *Convergence in distribution* means $F_{X_n}(x) \to F_X(x)$ wherever F_X is continuous.

The implication chain is pointwise $\Longrightarrow$ almost sure $\Longrightarrow$ in probability $\Longrightarrow$ in distribution. The reverse directions fail generally. Convergence in distribution specifies only the laws, not the joint relationship of X_n and X on a sample space. But if the distributional limit is a constant, convergence in distribution and convergence in probability are equivalent.

Example: $X_n(\omega) = \omega^n$ **on** $[0,1]$**.** For every $0 \leq \omega < 1$, $X_n(\omega) \to 0$, whereas at $\omega = 1$ the limit equals one. It therefore converges pointwise to $\mathbf{1}_{\{1\}}(\omega)$, almost surely to zero, in probability to zero and in distribution to a degenerate zero. For $0 < x < 1$, $F_{X_n}(x) = x^{1/n} \to 1$, consistent with a zero limit.

5.5 Weak and strong laws of large numbers

Suppose X_i are i.i.d. and $\mathbb{E}|X_1| < \infty$. The strong law states

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{a.s.} \mathbb{E}X_1.$$

A weaker consequence is $\bar{X}_n \xrightarrow{p} \mathbb{E}X_1$. Under finite variance, this follows directly from Chebyshev:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \to 0.$$

A typical proof of the strong law for bounded Bernoulli trials strengthens this tail bound using higher moments or exponential concentration and applies Borel–Cantelli. For general integrable variables, truncating extreme values reduces the problem to a bounded case; the remainder is handled using $\mathbb{E}|X_1| < \infty$. *The moment condition matters:* the sample mean of i.i.d. Cauchy observations does not settle to a finite population mean.

5.6 Central limit theorem and cumulant expansion

Let X_i be i.i.d. with finite mean μ and variance $0 < \sigma^2 < \infty$. Standardize $Z_i = (X_i - \mu)/\sigma$. When the MGF exists near zero, write $K(t) = \log \mathbb{E}[e^{tZ_1}]$. Since $K(0) = K'(0) = 0$ and $K''(0) = 1$, Taylor expansion yields

$$K(u) = \frac{1}{2}u^2 + o(u^2), \qquad \log \mathbb{E}\left[e^{t\sum_{i=1}^{n} Z_i/\sqrt{n}}\right] = nK(t/\sqrt{n}) \to \frac{t^2}{2}.$$

Therefore the standardized sum converges to $N(0,1)$. This MGF proof covers a useful subset of the usual CLT; the general finite-variance CLT does not require an MGF to exist in a neighborhood of zero.

Bernoulli case. For $X_i \sim \text{Bern}(p)$ and $p \in (0,1)$,

$$\frac{\sum X_i - np}{\sqrt{np(1-p)}} \Rightarrow N(0,1).$$

The approximation is useful when both np and $n(1-p)$ are sufficiently large. For a discrete tail approximated by a continuous normal distribution, a continuity correction often improves accuracy.

5.7 Delta method, Slutsky and the studentized mean

If $\sqrt{n}(T_n - \theta) \Rightarrow N(0, v)$ and g is differentiable at θ, the delta method gives

$$\sqrt{n}\{g(T_n) - g(\theta)\} \Rightarrow N\big(0, [g'(\theta)]^2 v\big).$$

Slutsky combines a distributional limit with another sequence converging in probability to a constant. Thus, for an i.i.d. sample with finite positive variance, the empirical standard deviation $S_n \xrightarrow{p} \sigma$ and

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \Rightarrow N(0,1).$$

For normal samples this statistic instead has the exact Student t_{n-1} law at finite n; as n grows, $t_{n-1} \Rightarrow N(0,1)$.

5.8 Monte Carlo integration and simulation error

To approximate $I = \int_0^1 f(x)\,dx$, generate independent $U_i \sim \text{Uniform}(0,1)$ and compute $\widehat{I}_n = n^{-1}\sum_i f(U_i)$. When $\mathbb{E}|f(U)| < \infty$, LLN guarantees $\widehat{I}_n \to I$ almost surely. When $\text{Var}(f(U)) = v < \infty$, CLT says

$$\sqrt{n}(\widehat{I}_n - I) \Rightarrow N(0, v), \qquad \text{SE}(\widehat{I}_n) \approx \sqrt{\widehat{v}/n}.$$

The error decreases at the rate $n^{-1/2}$, regardless of the dimension of an integrand when an appropriate sampling representation is available, though variance may grow considerably in harder problems.

Example: Gaussian integral. For $Z \sim N(0,1)$,

$$I = \int_0^1 \frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx = \mathbb{P}(0 < Z \le 1) = \Phi(1) - \frac{1}{2}.$$

One Monte Carlo method draws $U_i \sim \text{Uniform}(0,1)$ and averages $f(U_i) = e^{-U_i^2/2}/\sqrt{2\pi}$. Alternatively simulate $Z_i \sim N(0,1)$ and average $\mathbf{1}\{0 < Z_i \le 1\}$. Both are unbiased Monte Carlo approximations of the same integral, but their variances can differ.

5.9 Concentration inequalities and PAC error guarantees

For a generic sample mean error $R_n = \bar{X}_n - \mu$, a probabilistic accuracy statement has the form

$$\mathbb{P}(|R_n| \ge \varepsilon) \le \delta,$$

where ε is tolerance and δ is failure probability. Chebyshev gives $\mathbb{P}(|R_n| \ge \varepsilon) \le \sigma^2/(n\varepsilon^2)$. If $X_i \in [a,b]$ independently, Hoeffding yields

$$\mathbb{P}(|\bar{X}_n - \mu| \ge \varepsilon) \le 2\exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$

To ensure this upper bound is at most δ, it suffices to choose

$$n \ge \frac{(b-a)^2}{2\varepsilon^2}\log\frac{2}{\delta}.$$

Exponential tail bounds can be much tighter than the polynomial Chebyshev bound.

5.10 Chernoff method via exponential moments

For any random variable S with an exponential moment and $t > 0$, Markov's inequality applied to e^{tS} gives

$$\mathbb{P}(S \geq a) \leq e^{-ta}\mathbb{E}e^{tS}.$$

For $S_n = \sum_{i=1}^n X_i$ with i.i.d. X_i, $\mathbb{E}e^{tS_n} = [M_X(t)]^n$. Hence

$$\mathbb{P}(\bar{X}_n \geq a) \leq \inf_{t>0} \exp\{-n[ta - K_X(t)]\} = e^{-nI(a)},$$

where $K_X(t) = \log M_X(t)$ and $I(a) = \sup_{t>0}\{ta - K_X(t)\}$ is a one-sided Legendre transform. Lower tails use negative t. For Bernoulli variables, the resulting rate function is the binary relative entropy $D(a\|p)$. Thus concentration and the CLT arise from two approximations to the same log-MGF: global optimization for rare tails versus a local quadratic expansion for typical fluctuations.

5.11 Empirical distribution function

Given i.i.d. observations $X_1, \ldots, X_n$ with CDF F, the empirical CDF is

$$F_n(x) = \frac{1}{n}\sum_{i=1}^n \mathbf{1}\{X_i \leq x\}.$$

For fixed x, the indicators are Bernoulli with mean $F(x)$; therefore $\mathbb{E}F_n(x) = F(x)$ and $\mathrm{Var}(F_n(x)) = F(x)[1 - F(x)]/n$. The LLN gives pointwise consistency $F_n(x) \to F(x)$ almost surely for each fixed x. The empirical CDF is a direct example of how an unknown distribution can be learned through relative frequencies.

Key result

A three-level perspective: LLN gives consistency; CLT gives typical $n^{-1/2}$ fluctuations and approximate confidence intervals; concentration bounds give finite-sample probabilities of unusually large errors. These are different questions about the same estimator.

F–1 General CLT via cumulants and MGFs

Prompt. Give the MGF proof of the i.i.d. CLT assuming the MGF exists near 0. Let $Z_i = (X_i - \mu)/\sigma$ and $S_n = n^{-1/2}\sum_i Z_i$.
Sketch. With $K(u) = \log \mathbb{E}[e^{uZ_1}]$, $K(0) = K'(0) = 0$, $K''(0) = 1$. Taylor expansion gives $K(u) = u^2/2 + o(u^2)$. Independence now yields

$$\log M_{S_n}(t) = nK(t/\sqrt{n}) = t^2/2 + o(1), \qquad M_{S_n}(t) \to e^{t^2/2},$$

the standard normal MGF. Hence $S_n \xrightarrow{d} N(0,1)$.

Full MGF argument. First note that $M_{Z_1}(0) = 1$, $M'_{Z_1}(0) = \mathbb{E}Z_1 = 0$ and $M''_{Z_1}(0) = \mathbb{E}Z_1^2 = 1$. Differentiating $K = \log M$ gives

$$K'(u) = \frac{M'(u)}{M(u)}, \qquad K''(u) = \frac{M''(u)M(u) - M'(u)^2}{M(u)^2}.$$

Hence $K(0) = 0$, $K'(0) = 0$ and $K''(0) = 1$. Taylor's formula at zero gives $K(u) = u^2/2 + r(u)$, with $r(u)/u^2 \to 0$. Now

$$M_{n^{-1/2}\sum Z_i}(t) = \left[M_{Z_1}\left(\frac{t}{\sqrt{n}}\right)\right]^n,$$

so $\log M = t^2/2 + nr(t/\sqrt{n})$. The remainder is $nr(t/\sqrt{n}) = t^2\{r(t/\sqrt{n})/(t/\sqrt{n})^2\} \to 0$. Exponentiating yields $e^{t^2/2}$. The MGF convergence theorem gives convergence in distribution to the unique random variable with this MGF, the standard normal. This completes the proof under the near-zero MGF condition.

F–2 Bernoulli CLT as a Taylor-expansion example

Prompt. Prove the CLT for $X_i \overset{\text{iid}}{\sim} \text{Bern}(p)$ by the same MGF method, tracking the terms to second order.
Sketch. For $u = t/\sqrt{np(1-p)}$ and $Z_n = \sqrt{n}(\bar{X}_n - p)/\sqrt{p(1-p)}$,

$$\log M_{Z_n}(t) = n\{-pu + \log[1 + p(e^u - 1)]\}.$$

Use $e^u - 1 = u + u^2/2 + o(u^2)$ and $\log(1+v) = v - v^2/2 + o(v^2)$ to obtain $\log M_{Z_n}(t) = np(1-p)u^2/2 + o(1) = t^2/2 + o(1)$. Thus $Z_n \xrightarrow{d} N(0,1)$.

Track every second-order term. The Bernoulli MGF is $M_X(u) = q + pe^u$, where $q = 1 - p$. Set $u = t/\sqrt{npq}$ and $Z_n = (\sum X_i - np)/\sqrt{npq}$. Independence gives

$$\log \mathbb{E}e^{tZ_n} = -npu + n\log(1 + p(e^u - 1)).$$

Expand $e^u - 1 = u + u^2/2 + o(u^2)$ and put $v = p(e^u - 1) = pu + pu^2/2 + o(u^2)$. Then

$$\log(1+v) = v - v^2/2 + o(v^2) = pu + \frac{p(1-p)}{2}u^2 + o(u^2).$$

Subtracting npu cancels the terms of first order, leaving

$$\log \mathbb{E}e^{tZ_n} = \frac{npq}{2}u^2 + no(u^2) = \frac{t^2}{2} + o(1).$$

Therefore $Z_n \Rightarrow N(0,1)$. The cancellation of the linear term is the mathematical reason for centering by np.

F–3 A random ratio with a gamma limit

Prompt. For $\xi_i \overset{\text{iid}}{\sim} N(0,1)$ set $F_n = (\xi_1^2 + \xi_2^2)/(\sum_{i=1}^n \xi_i^2)$. Determine the limit law of $2nF_n$.
Sketch. $\sum_i \xi_i^2/n \xrightarrow{a.s.} 1$ by the LLN, while $\xi_1^2 + \xi_2^2 \sim \chi_2^2$ for every $n \geq 2$. Consequently

$$2nF_n = \frac{2(\xi_1^2 + \xi_2^2)}{n^{-1}\sum_{i=1}^n \xi_i^2} \xrightarrow{d} 2\chi_2^2 = \text{Gamma}(1, \text{scale } 4).$$

Important: The factor $2n$ must be retained when taking the limit. A tempting calculation states Gamma$(1, \text{scale } 2)$ but drops a factor of two in its algebra. See the clarification box below.

Exact distribution check. The numerator $A = \xi_1^2 + \xi_2^2 \sim \chi_2^2$ and the remaining sum $B = \sum_{i=3}^n \xi_i^2 \sim \chi_{n-2}^2$ are independent. Therefore

$$F_n = \frac{A}{A+B} \sim \text{Beta}\left(1, \frac{n-2}{2}\right) \quad (n > 2).$$

The denominator of $2nF_n$ is the entire sum. By the SLLN $n^{-1}(A+B) \to 1$ almost surely, whereas A does not change its distribution with n. Slutsky gives $2nF_n \Rightarrow 2A$. Since A has density $e^{-x/2}/2$ on $x > 0$, the random variable $2A$ has density $e^{-y/4}/4$. This is exponential with mean 4, equivalently Gamma(shape 1, scale 4). Computing the limiting mean from the exact beta law also checks the scale: $\mathbb{E}[2nF_n] = 2n \cdot (2/n) = 4$.

Scaling check: Since $F_n \sim \text{Beta}(1, (n-2)/2)$ for $n > 2$, $nF_n \xrightarrow{d} \chi_2^2$ and $2nF_n \xrightarrow{d} 2\chi_2^2$; a gamma(shape 1, scale 4) limit has mean 4. The alternative expression corresponds to scale 2, corresponding instead to nF_n. The corrected factor is explicitly indicated here rather than silently changing the exercise.

F–5 Self-normalized sample mean

Prompt. If X_i are i.i.d. with $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = 1$, find the limit of $T_n = \sqrt{n}\bar{X}_n / \sqrt{n^{-1}\sum X_i^2}$.

Sketch. CLT: $\sqrt{n}\bar{X}_n \xrightarrow{d} N(0,1)$. SLLN: $n^{-1}\sum X_i^2 \xrightarrow{a.s.} 1$. Continuous mapping makes the denominator converge to 1 in probability; Slutsky then gives $T_n \xrightarrow{d} N(0,1)$.

A precise three-step Slutsky proof. (1) The ordinary CLT yields $\sqrt{n}\bar{X}_n \Rightarrow Z$ where $Z \sim N(0,1)$. (2) Because $\mathbb{E}X_i^2 = 1$, the strong law applied to X_i^2 gives $V_n = n^{-1}\sum X_i^2 \to 1$ almost surely. (3) By continuity of the positive square root, $\sqrt{V_n} \to 1$ in probability. Slutsky then implies $\sqrt{n}\bar{X}_n / \sqrt{V_n} \Rightarrow Z$. Notice that independence between numerator and denominator is not required for Slutsky, since the denominator converges to a constant.

PART II

Statistical Inference

6 Estimation, Sufficiency and Fisher Information

Estimation translates random measurements into information about unknown parameters. The main tools are moment matching, likelihood maximization, sufficiency, conditional estimation and information bounds.

6.1 Methods at a glance

For i.i.d. data, likelihood $L(\theta) = \prod_i f_\theta(X_i)$; maximize $\ell(\theta) = \log L(\theta)$ respecting constraints/support. The *method of moments* equates empirical and population moments. T is sufficient for θ if the conditional law of the sample given T is free of θ (or via factorization). If T is complete sufficient and U is unbiased for $g(\theta)$, $\mathbb{E}[U \mid T]$ is the UMVU estimator. Fisher information and the Cramér–Rao bound:

$$I_n(\theta) = \mathbb{E}_\theta\left[\left(\frac{\partial}{\partial\theta}\ell(\theta)\right)^2\right], \qquad \mathrm{Var}_\theta(\widehat{g}) \geq \frac{\{g'(\theta)\}^2}{I_n(\theta)}$$

for unbiased estimators under regularity assumptions.

6.2 Statistical model, estimator, bias and risk

A parametric model is a family of probability laws $\{\mathbb{P}_\theta : \theta \in \Theta\}$. The parameter θ is fixed but unknown; the observed sample is random under $\mathbb{P}_\theta$. An estimator $\delta(X_1, \ldots, X_n)$ is a statistic used to estimate some target $g(\theta)$. Its bias, variance and squared-error risk are

$$\mathrm{Bias}_\theta(\delta) = \mathbb{E}_\theta\delta - g(\theta), \quad \mathrm{MSE}_\theta(\delta) = \mathrm{Var}_\theta(\delta) + \mathrm{Bias}_\theta(\delta)^2.$$

Unbiasedness means $\mathbb{E}_\theta\delta = g(\theta)$ for all parameter values; consistency means $\delta_n \xrightarrow{p} g(\theta)$ for each fixed θ. Neither unbiasedness nor consistency guarantees uniformly small finite-sample mean-squared error.

6.3 Method of moments and maximum likelihood

The method of moments chooses $\widehat{\theta}$ so that empirical moments agree with corresponding population moments, for example $\bar{X} = m_1(\theta) = \mathbb{E}_\theta X$. With a vector parameter, use enough moments to identify every component. The method is often algebraically simple, but efficiency is not guaranteed.

For independent data with density or mass f_θ, maximum likelihood chooses a maximizer of

$$L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^n f_\theta(x_i), \qquad \ell(\theta) = \sum_{i=1}^n \log f_\theta(x_i).$$

When support does not depend on θ and the optimum is interior, solve $\ell'(\theta) = 0$ and check a maximum. If the distribution's support depends on θ, include the indicator restrictions: the maximum is often at a sample minimum or maximum. For $X_i \sim \mathrm{Uniform}(0, \theta)$, $\widehat{\theta} = X_{(n)}$ because $L(\theta) = \theta^{-n}\mathbf{1}\{\theta \geq X_{(n)}\}$.

Invariance of MLE. For a one-to-one transformation $\eta = g(\theta)$, the MLE is $\widehat{\eta} = g(\widehat{\theta})$; appropriate maximizer conventions also apply for non-injective g. This is an optimization property, not a statement about unbiasedness.

6.4 Sufficiency, factorization and completeness

A statistic $T(X)$ is sufficient if the conditional distribution of the complete sample given T does not depend on θ. The factorization criterion is

$$f_\theta(x_1,\dots,x_n) = h(x_1,\dots,x_n)g_\theta(T(x_1,\dots,x_n)),$$

where h does not depend on θ. Thus all dependence on the parameter appears through T. Examples include $\sum_i X_i$ for iid Poisson or exponential rates and $X_{(n)}$ for a uniform endpoint. The normal location-scale family is summarized by $(\sum_i X_i, \sum_i X_i^2)$.

A family of distributions for T is complete if $\mathbb{E}_\theta h(T) = 0$ for every θ implies $h(T) = 0$ almost surely under every parameter value. Completeness is crucial for proving uniqueness of unbiased functions of a sufficient statistic. If U is unbiased for $g(\theta)$, then $\mathbb{E}[U \mid T]$ is still unbiased and has no larger variance, by the law of total variance; this is the Rao–Blackwell theorem. When T is complete and sufficient, the resulting unbiased function of T is the unique UMVU estimator (Lehmann–Scheffé principle).

Poisson illustration. With $T = \sum_i X_i \sim \text{Poisson}(n\theta)$,

$$\mathbb{E}[X_1 \mid T] = T/n, \qquad X_1 \mid T = t \sim \text{Bin}(t, 1/n).$$

The unconditional estimator $2X_1$ and its conditional version $2T/n$ are both unbiased for 2θ, but the latter removes conditional noise and is UMVU.

6.5 Score, information and the Cramér–Rao lower bound

For one parameter, the score is $U(\theta) = \partial\ell(\theta)/\partial\theta$. Under the regularity conditions permitting differentiation inside an integral,

$$\mathbb{E}_\theta U(\theta) = 0, \qquad I_n(\theta) = \text{Var}_\theta\, U(\theta) = -\mathbb{E}_\theta \ell''(\theta).$$

For iid data, $I_n(\theta) = nI_1(\theta)$. An unbiased estimator δ of $g(\theta)$ satisfies

$$\text{Var}_\theta(\delta) \geq \frac{[g'(\theta)]^2}{I_n(\theta)}.$$

The derivation differentiates $\mathbb{E}_\theta\delta = g(\theta)$ to obtain $\text{Cov}(\delta, U) = g'(\theta)$ and applies Cauchy–Schwarz. The bound need not apply unchanged to models with parameter-dependent support or without integrability conditions.

For a Poisson sample, $\ell(\theta) = -n\theta + T\log\theta + \text{const}$, so $U = T/\theta - n$ and $I_n(\theta) = n/\theta$. A target $g(\theta) = \theta^2 e^{-\theta}$ has $g'(\theta) = \theta(2-\theta)e^{-\theta}$; the CRLB is $\theta^3(2-\theta)^2 e^{-2\theta}/n$.

6.6 Asymptotics of estimators

For a regular identifiable model with nonsingular Fisher information, the maximum-likelihood estimator is often asymptotically normal:

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \Rightarrow N\big(0, I_1(\theta_0)^{-1}\big).$$

The claim depends on regularity and is not automatic for endpoint families such as Uniform$(0,\theta)$; there the maximum order statistic converges to θ at the rate n rather than the usual $\sqrt{n}$. For a smooth transformed parameter $g(\theta)$, asymptotic variance follows from the delta method.

Key result

Estimator workflow: include the parameter-dependent support; find MoM/MLE; check bias and consistency; identify sufficient statistics; improve unbiased estimators by conditioning; compute Fisher information only when regularity holds; examine the actual limiting scale.

P2–2 Power density, transformation, MoM and MLE

Prompt. $f_\theta(x) = \theta x^{\theta-1}$, $0 < x < 1$, $\theta > 0$. (a) Find density of $Y = -\log X$; (b) find method-of-moments estimate; (c) find the MLE.
Answers. $F_X(x) = x^\theta$; hence $Y \sim$ Exp(rate θ), $f_Y(y) = \theta e^{-\theta y}$, $y > 0$. Since $\mathbb{E}X = \theta/(\theta+1)$,

$$\widehat{\theta}_{\text{MoM}} = \frac{\bar{X}}{1-\bar{X}}.$$

The log likelihood is $\ell(\theta) = n\log\theta + (\theta-1)\sum_i \log x_i$; its score is $n/\theta + \sum_i \log x_i$, so

$$\widehat{\theta}_{\text{MLE}} = -\frac{n}{\sum_{i=1}^n \log X_i}, \qquad \ell''(\theta) = -n/\theta^2 < 0.$$

Transformation by two independent methods. The inverse of $Y = -\log X$ is $X = e^{-Y}$ and $|dx/dy| = e^{-y}$. Hence

$$f_Y(y) = \theta(e^{-y})^{\theta-1}e^{-y} = \theta e^{-\theta y}, \quad y > 0.$$

Equivalently, $\mathbb{P}(Y \le y) = \mathbb{P}(X \ge e^{-y}) = 1 - (e^{-y})^\theta = 1 - e^{-\theta y}$. Both calculations give the same exponential distribution. Since $\mathbb{E}\log X = -1/\theta$ and $\text{Var}(\log X) = 1/\theta^2$, the MLE can also be written $\widehat{\theta} = 1/\overline{(-\log X)}$.
Method of moments. $\int_0^1 x\theta x^{\theta-1}dx = \theta/(\theta+1)$, so $\bar{X}(\theta+1) = \theta$. Solve for θ to get $\bar{X}/(1-\bar{X})$. This is not generally identical to the MLE, because moment matching and likelihood maximization solve different optimization problems.
Likelihood check. $\ell'(\theta) = n/\theta + \sum \log x_i$ vanishes once on $\theta > 0$ because every $\log x_i < 0$. The second derivative $-n/\theta^2$ is strictly negative, confirming a unique global maximizer.

P2–3 (estimation parts) Uniform endpoint

Prompt. For $X_i \overset{\text{iid}}{\sim} \text{Uniform}(0,\theta)$, identify MLE, its bias and a sufficient statistic; explain asymptotic normality by a normal-mean contrast.
Answers. $L(\theta) = \theta^{-n}\mathbf{1}\{\theta \ge X_{(n)}\}$ with $X_{(n)} = \max_i X_i$, therefore $\widehat{\theta} = X_{(n)}$ and $X_{(n)}$ is sufficient. Since $\mathbb{P}(X_{(n)} \le t) = (t/\theta)^n$ for $0 < t < \theta$,

$$\mathbb{E}[X_{(n)}] = \frac{n}{n+1}\theta, \quad \text{Bias}(\widehat{\theta}) = -\frac{\theta}{n+1}.$$

It is consistent but not unbiased. In contrast, for $X_i \overset{\text{iid}}{\sim} N(\mu,1)$, $\widehat{\mu} = \bar{X}$ and $\sqrt{n}(\bar{X}-\mu) \sim N(0,1)$ exactly.
The UMP-test part of P2–3 is completed in §8, without repeating this derivation.

Distribution, bias and consistency in detail. For $M = X_{(n)}$,

$$\mathbb{P}(M \le m) = \prod_{i=1}^n \mathbb{P}(X_i \le m) = (m/\theta)^n, \quad 0 < m < \theta.$$

Therefore $f_M(m) = nm^{n-1}/\theta^n$, $\mathbb{E}M = \int_0^\theta m f_M(m)dm = n\theta/(n+1)$ and $\text{Var}(M) = n\theta^2/[(n+1)^2(n+2)]$. Its bias is $-\theta/(n+1)$. The unbiased correction is $\widetilde{\theta} = (n+1)M/n$. Also, for any fixed $\varepsilon \in (0,\theta)$,

$$\mathbb{P}(|M-\theta| > \varepsilon) = \mathbb{P}(M < \theta - \varepsilon) = \left(1 - \frac{\varepsilon}{\theta}\right)^n \to 0.$$

Thus M is consistent. The asymptotic scaling is non-Gaussian: for $t \ge 0$, $\mathbb{P}(n(\theta-M)/\theta > t) = (1 - t/n)^n \to e^{-t}$. The nonregular endpoint model therefore cannot be treated by the standard interior-MLE normal approximation.

P2–5 Poisson sufficiency, Rao–Blackwell and CRLB

Prompt. For $X_i \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$: (a) find a complete sufficient statistic; (b) find UMVU estimator of 2θ; (c) show $2\mathbf{1}\{X_1 = 2\}$ is unbiased for $g(\theta) = \theta^2 e^{-\theta}$; (d) compute its CRLB.
Answers. $T = \sum X_i \sim \text{Poisson}(n\theta)$ is complete sufficient. Since $X_1 \mid T \sim \text{Bin}(T, 1/n)$,

$$\mathbb{E}[2X_1 \mid T] = \frac{2T}{n} \quad \text{(UMVU for } 2\theta\text{)}.$$

$\mathbb{E}[2\mathbf{1}\{X_1 = 2\}] = 2e^{-\theta}\theta^2/2! = \theta^2 e^{-\theta}$. Using $g'(\theta) = \theta(2-\theta)e^{-\theta}$ and $I_n(\theta) = n/\theta$,

$$\text{CRLB}(g) = \frac{e^{-2\theta}\theta^3(2-\theta)^2}{n}.$$

Factorization and completeness. The joint Poisson likelihood is

$$L(\theta; x) = \frac{e^{-n\theta}\theta^{\sum x_i}}{\prod_i x_i!}, \qquad T = \sum_i X_i \sim \text{Poisson}(n\theta).$$

The factorization theorem makes T sufficient. If $\mathbb{E}_\theta h(T) = 0$ for all $\theta > 0$, then

$$0 = e^{-n\theta} \sum_{t=0}^{\infty} h(t) \frac{(n\theta)^t}{t!}.$$

The power series vanishes for every positive θ, so each coefficient must vanish: $h(t) = 0$ for all integers $t \geq 0$ with positive probability. Thus T is complete.
Rao–Blackwell calculation. For the unbiased estimator $2X_1$, use $X_1 \mid T = t \sim \text{Bin}(t, 1/n)$, giving $\mathbb{E}(2X_1 \mid T) = 2T/n$. Its variance is $4\,\text{Var}(T)/n^2 = 4\theta/n$. Any other unbiased estimator cannot improve its variance within the full class under the completeness/sufficiency conditions.
Lower bound for the nonlinear target. Because $U(\theta) = T/\theta - n$ and $\text{Var}\, U = n/\theta$, the information is $I_n = n/\theta$. For $g(\theta) = \theta^2 e^{-\theta}$ the CRLB is $[\theta(2-\theta)e^{-\theta}]^2/(n/\theta)$. The estimator $2\mathbf{1}\{X_1 = 2\}$ is unbiased but is not an estimator attaining the bound in general.

P1–1 Exact laws of normal sample variance and SD

Prompt. For $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, define $U^2 = \frac{1}{n-1}\sum_i (X_i - \bar{X})^2$, $U = \sqrt{U^2}$. (a) $\text{Var}(U^2)$; (b) density of U; (c) $\mathbb{E}U$; (d) prove $\mathbb{E}U \leq \sigma$.
Answers. Since $Q = (n-1)U^2/\sigma^2 \sim \chi^2_{n-1}$,

$$\text{Var}(U^2) = \frac{2\sigma^4}{n-1}, \qquad f_U(u) = \frac{2u^{n-2}e^{-(n-1)u^2/(2\sigma^2)}}{\Gamma((n-1)/2)\{2\sigma^2/(n-1)\}^{(n-1)/2}}, \quad u > 0.$$

Using $\mathbb{E}Q^{1/2} = \sqrt{2}\Gamma(n/2)/\Gamma((n-1)/2)$,

$$\mathbb{E}U = \frac{\sigma\sqrt{2}}{\sqrt{n-1}} \frac{\Gamma(n/2)}{\Gamma((n-1)/2)} \leq \sqrt{\mathbb{E}(U^2)} = \sigma.$$

The inequality is Jensen applied to the concave square root.

Density derivation. Write $k = n-1$ and $Q = kU^2/\sigma^2 \sim \chi^2_k$ with density $f_Q(q) = q^{k/2-1}e^{-q/2}/[2^{k/2}\Gamma(k/2)]$. For $u > 0$, the transformation is $q = ku^2/\sigma^2$, $dq/du = 2ku/\sigma^2$, so

$$f_U(u) = \frac{2(k/\sigma^2)^{k/2}}{2^{k/2}\Gamma(k/2)} u^{k-1} \exp\left(-\frac{ku^2}{2\sigma^2}\right), \quad u > 0.$$

This is the scale-adjusted chi distribution, not a chi-square density. Since $\text{Var}(Q) = 2k$, $\text{Var}(U^2) = \sigma^4\,\text{Var}(Q)/k^2 = 2\sigma^4/k$.

Gamma expectation. $\mathbb{E}\sqrt{Q} = 2^{1/2}\Gamma((k+1)/2)/\Gamma(k/2)$, giving $\mathbb{E}U = (\sigma/\sqrt{k})\sqrt{2}\Gamma((k+1)/2)/\Gamma(k/2)$. Jensen on $\sqrt{\cdot}$ yields $\mathbb{E}U = \mathbb{E}\sqrt{U^2} \le \sqrt{\mathbb{E}U^2} = \sigma$. Hence U^2 is unbiased for σ^2, but U itself is biased downward for σ at finite n.

P1–3 MLE from signs only, with consistency

Prompt. $X_i \overset{\text{iid}}{\sim} N(\mu, 1)$ but only $Y_i = \mathbf{1}\{X_i \ge 0\}$ are observed. (a) Find MLE of μ; (b) show convergence in probability.

Answers. $Y_i \overset{\text{iid}}{\sim} \text{Bern}(\Phi(\mu))$. Maximizing $\sum_i [Y_i \log \Phi(\mu) + (1 - Y_i)\log(1 - \Phi(\mu))]$ yields

$$\widehat{\mu} = \Phi^{-1}(\bar{Y}), \qquad 0 < \bar{Y} < 1.$$

For $\bar{Y} = 0$ or 1, the likelihood supremum occurs at $\mu \to -\infty$ or $+\infty$, respectively (no finite MLE). By LLN $\bar{Y} \xrightarrow{p} \Phi(\mu)$, and the continuous mapping theorem gives $\widehat{\mu} \xrightarrow{p} \mu$ for each finite true μ.

Log-likelihood and derivative. Observing only the signs is equivalent to observing $Y_i \sim \text{Bern}(p)$ with $p = \Phi(\mu)$. If $S = \sum_i Y_i$, the likelihood is

$$L(\mu) = \Phi(\mu)^S[1 - \Phi(\mu)]^{n-S}, \qquad \ell'(\mu) = \phi(\mu)\left[\frac{S}{\Phi(\mu)} - \frac{n-S}{1-\Phi(\mu)}\right].$$

For $0 < S < n$, the first-order condition reduces to $S - n\Phi(\mu) = 0$, giving $\widehat{\mu} = \Phi^{-1}(S/n)$. If $S = 0$, the supremum is at $\mu \downarrow -\infty$; if $S = n$, it is at $\mu \uparrow +\infty$. These boundary events become improbable as n grows when the true μ is finite.

Consistency. For the true parameter, $p = \Phi(\mu) \in (0, 1)$. SLLN gives $S/n \to p$ almost surely. Since Φ^{-1} is continuous at p, the continuous mapping theorem yields $\widehat{\mu} \to \mu$ on the interior event and hence convergence in probability. The estimator can be studied further by the delta method: $\sqrt{n}(\widehat{\mu} - \mu)$ has asymptotic variance $p(1-p)/\phi(\mu)^2$.

P1–5 Two exponentials: MLE and risk pitfall

Prompt. Independent $X \sim \text{Exp(rate } \theta)$ and $Y \sim \text{Exp(rate } 1/\theta)$. (a) Find MLE of θ; (b) find c so that $c\widehat{\theta}$ is unbiased; (c) construct a better unbiased estimator in MSE for every $\theta > 0$.

Answers. $L(\theta) = e^{-\theta x - y/\theta}$ and $\ell'(\theta) = -x + y/\theta^2$, so $\widehat{\theta}_{\text{MLE}} = \sqrt{Y/X}$. With $U = \theta X, V = Y/\theta$ independent Exp(1),

$$\mathbb{E}\sqrt{Y/X} = \theta\Gamma(1/2)\Gamma(3/2) = \frac{\pi\theta}{2}.$$

Thus $\widehat{\theta}_U = (2/\pi)\sqrt{Y/X}$ is unbiased. But $\mathbb{E}[\widehat{\theta}_U^2] = (4/\pi^2)\mathbb{E}[Y]\mathbb{E}[1/X] = \infty$, since $\mathbb{E}[1/X] = \infty$. The unbiased $\delta = Y$ has $\text{Var}(Y) = \theta^2 < \infty$, so it strictly improves the MSE.

Maximum likelihood step by step. The independent densities are $f_X(x) = \theta e^{-\theta x}$ and $f_Y(y) = \theta^{-1}e^{-y/\theta}$, for $x, y > 0$. Multiplication cancels the factors θ and θ^{-1}:

$$L(\theta) = e^{-\theta x - y/\theta}, \quad \ell'(\theta) = -x + y/\theta^2, \quad \ell''(\theta) = -2y/\theta^3 < 0.$$

Thus the sole critical point is the maximum $\widehat{\theta}_{ML} = \sqrt{y/x}$.

Unbiased correction. Let $U = \theta X$ and $V = Y/\theta$. Each has Exp(rate 1), and they are independent. Then $\widehat{\theta}_{ML} = \theta\sqrt{V/U}$. Gamma integration gives

$$\mathbb{E}V^{1/2} = \Gamma(3/2) = \frac{\sqrt{\pi}}{2}, \qquad \mathbb{E}U^{-1/2} = \Gamma(1/2) = \sqrt{\pi}.$$

Thus $\mathbb{E}\widehat{\theta}_{ML} = \pi\theta/2$, so $2\widehat{\theta}_{ML}/\pi$ is unbiased.

Why unbiased does not mean good. Its second moment is

$$\mathbb{E}\left(\frac{2}{\pi}\sqrt{Y/X}\right)^2 = \frac{4}{\pi^2}\mathbb{E}Y\,\mathbb{E}(1/X) = \infty,$$

because $\mathbb{E}(1/X) = \int_0^\infty x^{-1}\theta e^{-\theta x}dx = \infty$ near zero. The estimator $\delta(X,Y) = Y$ is unbiased ($\mathbb{E}Y = \theta$) and has variance θ^2, so it has strictly smaller finite risk for every $\theta > 0$.

7 Confidence Intervals and Pivotal Quantities

A $(1-\alpha)$ confidence interval is a random set $C(X)$ with frequentist coverage $\mathbb{P}_\theta\{\theta \in C(X)\} \geq 1-\alpha$. A *pivot* has a known sampling distribution independent of unknown parameters; invert its quantiles to obtain an interval. For discrete data, exact coverage is often conservative.

7.1 Coverage, confidence level and interval interpretation

A random interval $[L(X), U(X)]$ has confidence level at least $1-\alpha$ when $\mathbb{P}_\theta\{L(X) \leq \theta \leq U(X)\} \geq 1-\alpha$ for every θ in the model. The endpoints are random before data collection; the parameter is not random in the usual frequentist model. After observing data, the reported numerical interval is fixed. "90% confidence" refers to the long-run coverage of the interval-producing procedure.

A parameter-free pivot $Q(X,\theta)$ with known distribution enables interval construction. Choose constants a, b such that $\mathbb{P}(a \leq Q(X,\theta) \leq b) = 1-\alpha$, then solve the two inequalities for θ. A monotone transformation of a pivot is another pivot if its sampling law is known.

7.2 Normal and chi-square pivotal families

If $X_i \sim N(\mu, \sigma^2)$ iid with σ known,

$$\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \sim N(0,1), \qquad \mu \in \left[\bar{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right].$$

With σ unknown, let $S^2 = (n-1)^{-1}\sum(X_i - \bar{X})^2$. Then $(\bar{X}-\mu)/(S/\sqrt{n}) \sim t_{n-1}$ and $((n-1)S^2/\sigma^2) \sim \chi^2_{n-1}$. These yield exact normal-theory intervals for μ and σ^2, respectively.

7.3 Exact intervals for Poisson means and shifted endpoints

If $Y \sim \text{Poisson}(n\lambda)$ is observed as y, Poisson–gamma waiting-time duality gives a convenient exact interval, often called the Garwood interval:

$$\lambda \in \left[\frac{\chi^2_{2y,\alpha/2}}{2n},\ \frac{\chi^2_{2(y+1),1-\alpha/2}}{2n}\right],$$

with the lower endpoint set to zero when $y = 0$. This interval has coverage at least $1-\alpha$, generally greater for some λ because counts are discrete. The notation $\chi^2_{\nu,q}$ means the lower-tail q-quantile. A shifted exponential sample minimum provides an exact exponential pivot for its unknown endpoint; intervals follow by inverting its two tails.

7.4 Shortest intervals and the role of symmetry

When a symmetric unimodal model admits a location pivot, a central interval is often shortest among intervals with a given coverage. For Laplace density $\frac{1}{2}e^{-|x-\theta|}$, the two-tail probability outside $[X-b, X+c]$ equals $\frac{1}{2}(e^{-b}+e^{-c})$. Fixing $e^{-b}+e^{-c} = 2\alpha$ and minimizing $b+c$ gives $b = c = -\log\alpha$. The key is not merely a "center the interval" heuristic, but a constrained optimization using the actual tail law.

P1–2 Shortest Laplace location interval

Prompt. If $f(x \mid \theta) = \frac{1}{2}e^{-|x-\theta|}$, find $\mathbb{P}(\theta - c \le X \le \theta + b)$ for $b, c > 0$, then choose b, c to minimize length of $[X - b, X + c]$ at coverage $1 - \alpha$.
Answers. The event has probability

$$\mathbb{P}(\theta - c \le X \le \theta + b) = 1 - \frac{1}{2}(e^{-b} + e^{-c}).$$

Coverage requires $e^{-b} + e^{-c} = 2\alpha$. By symmetry / Lagrange multipliers, $b = c = -\log\alpha$, giving shortest length $-2\log\alpha$ and interval

$$\boxed{[X + \log\alpha,\ X - \log\alpha]}.$$

Tail integration and optimization. Since the Laplace density has one exponential tail on each side of θ,

$$\mathbb{P}(\theta - c \le X \le \theta + b) = \frac{1}{2}\int_{-c}^{0} e^{t}dt + \frac{1}{2}\int_{0}^{b} e^{-t}dt = 1 - \frac{1}{2}(e^{-c} + e^{-b}).$$

Requiring coverage $1 - \alpha$ gives $e^{-b} + e^{-c} = 2\alpha$. Minimize $d = b + c$ subject to that constraint. With $\mathcal{L} = b + c + \lambda(e^{-b} + e^{-c} - 2\alpha)$, the first-order conditions are $1 - \lambda e^{-b} = 0$, $1 - \lambda e^{-c} = 0$; hence $b = c$. The constraint then gives $e^{-b} = \alpha$ and $b = c = -\log\alpha$. Convexity of $x \mapsto e^{-x}$ or a second-order check confirms the optimum. The data-centered interval is $[X + \log\alpha, X - \log\alpha]$, with length $-2\log\alpha$.

F–8 Exact confidence interval for a Poisson mean

Prompt. Derive Poisson–chi-square tail identities from the Poisson process and gamma waiting times; construct a $(1 - \alpha)$ equal-tailed interval for λ_0 when $Y = \sum_{i=1}^{n} X_i \sim \text{Poisson}(n\lambda_0)$ with observed total y; evaluate the numerical case $n = 10, y = 6, \alpha = .1$.
Steps. In a Poisson process, $\{N(1) \le y\} = \{W_{y+1} > 1\}$ and $\{N(1) \ge y\} = \{W_y \le 1\}$. Since $2\lambda W_k \sim \chi^2_{2k}$,

$$\mathbb{P}(\text{Poisson}(\lambda) \le y) = \mathbb{P}(\chi^2_{2(y+1)} > 2\lambda), \quad \mathbb{P}(\text{Poisson}(\lambda) \ge y) = \mathbb{P}(\chi^2_{2y} \le 2\lambda).$$

Inverting the tails gives the standard Garwood interval

$$\boxed{\lambda_0 \in \left[\frac{\chi^2_{2y,\alpha/2}}{2n},\ \frac{\chi^2_{2(y+1),\,1-\alpha/2}}{2n}\right]}, \qquad y \ge 1.$$

For $y = 0$ the lower endpoint is 0. In the example, $\chi^2_{12,.05} \approx 5.226$ and $\chi^2_{14,.95} \approx 23.685$, yielding $[0.2613, 1.1843]$ (rounding convention may change the last decimal).

Poisson–gamma identity in both directions. Let $N(1) \sim \text{Poisson}(\lambda)$ be the count in unit time and G_k the time of the kth event. By duality, $\{N(1) \le y\} = \{G_{y+1} > 1\}$ and $\{N(1) \ge y\} = \{G_y \le 1\}$. With $2\lambda G_k \sim \chi^2_{2k}$,

$$\mathbb{P}_\lambda\{N(1) \le y\} = \mathbb{P}\{\chi^2_{2(y+1)} > 2\lambda\},$$
$$\mathbb{P}_\lambda\{N(1) \ge y\} = \mathbb{P}\{\chi^2_{2y} \le 2\lambda\}, \qquad y \ge 1.$$

To invert the first identity for an upper bound, solve the appropriate lower-tail equation for 2λ at the $1 - \alpha/2$ quantile of $\chi^2_{2(y+1)}$. To obtain the lower bound use the $\alpha/2$ quantile of χ^2_{2y}. If the total count is $Y = \sum_{j=1}^{n} X_j \sim \text{Poisson}(n\lambda_0)$, divide both bounds for the total intensity by n.
Numerical inversion. For $n = 10$, observed count $y = 6$ and confidence 0.90, the degrees of freedom are 12 and 14, with $\chi^2_{12,.05} \approx 5.226$ and $\chi^2_{14,.95} \approx 23.685$. Thus $\lambda_0 \in [5.226/20, 23.685/20] \approx [0.2613, 1.1843]$. For $y = 0$ there is no strictly positive lower bound of this form, and the interval begins at zero.

Quantile convention: Lower-tail quantiles must be used consistently: the lower endpoint uses the $\alpha/2$ quantile; the upper endpoint uses the $1-\alpha/2$ quantile.

F–9 Shifted exponential endpoint interval

Prompt. For i.i.d. $f(x \mid \mu) = e^{-(x-\mu)}\mathbf{1}\{x \geq \mu\}$, use $Y = X_{(1)}$ to construct a two-sided $(1-\alpha)$ equal-tailed confidence interval for the location μ.
Answers. $Y-\mu \sim$ Exp(rate n). If $q_L = -\log(1-\alpha/2)/n$ and $q_U = -\log(\alpha/2)/n$ are its central quantiles, then

$$\mathbb{P}(q_L \leq Y - \mu \leq q_U) = 1-\alpha.$$

Solving for μ gives

$$\boxed{\mu \in \left[Y + \frac{1}{n}\log(\alpha/2), \quad Y + \frac{1}{n}\log(1-\alpha/2)\right]}.$$

Here Y is the sufficient sample minimum; the interval uses a parameter-free pivot.

Order-statistic distribution. If $X_i = \mu + E_i$ with independent $E_i \sim \text{Exp}(1)$, then $Y = \min_i X_i = \mu + \min_i E_i$. For $t \geq 0$,

$$\mathbb{P}(Y - \mu > t) = \prod_{i=1}^{n} \mathbb{P}(E_i > t) = e^{-nt}.$$

Hence $Y - \mu \sim$ Exp(rate n), with CDF $1-e^{-nt}$. To give equal tail probability $\alpha/2$ at each end, take q_L and q_U satisfying $1-e^{-nq_L} = \alpha/2$ and $e^{-nq_U} = \alpha/2$. Thus

$$q_L = -\frac{1}{n}\log(1-\alpha/2), \qquad q_U = -\frac{1}{n}\log(\alpha/2).$$

The event $q_L \leq Y - \mu \leq q_U$ is equivalent to $Y - q_U \leq \mu \leq Y - q_L$, giving exactly the displayed interval. There is no approximation: the coverage is $1-\alpha$ under the stated exponential model.

8 Hypothesis Tests, NP Lemma and LRTs

8.1 Testing notation and test-statistic construction

A test $\phi(\boldsymbol{X}) \in [0,1]$ rejects with probability ϕ. Its *power function* is $\gamma(\theta) = \mathbb{E}_\theta \phi$, and its *size* is $\sup_{\theta \in \Theta_0} \gamma(\theta)$. A Type II error at $\theta \in \Theta_1$ equals $1-\gamma(\theta)$. For simple $H_0 : \theta = \theta_0$ versus simple $H_1 : \theta = \theta_1$, the Neyman–Pearson lemma rejects for *large* $L(\theta_1)/L(\theta_0)$. A UMP test has maximal power for every alternative at a fixed level.

For composite hypotheses the generalized likelihood ratio is

$$\Lambda(\boldsymbol{x}) = \frac{\sup_{\theta \in \Theta_0} L(\theta; \boldsymbol{x})}{\sup_{\theta \in \Theta} L(\theta; \boldsymbol{x})} \in [0,1],$$

and one rejects for *small* Λ (or large $-2\log\Lambda$). Under suitable smooth regularity conditions, $-2\log\Lambda \xrightarrow{d} \chi^2_r$ for r restrictions (Wilks' theorem).

8.2 Definitions: null, alternative, error probabilities and power

A nonrandomized test is an indicator $\phi(X) \in \{0,1\}$, and a randomized test may take values in $[0,1]$. With null parameter set Θ_0 and alternative Θ_1, the power function is $\gamma(\theta) = \mathbb{E}_\theta \phi(X)$. Its size is $\sup_{\theta \in \Theta_0} \gamma(\theta)$. For simple $H_0 : \theta = \theta_0$, Type I error is $\gamma(\theta_0)$ and Type II error at θ_1 is $1-\gamma(\theta_1)$.

For a fixed test statistic T and a rejection set R, the p-value is a null tail probability of outcomes at least as incompatible with H_0 as the observed data, under a specified ordering. The p-value is not the probability

that H_0 is true. Exact tail rules matter in discrete and two-sided problems.

8.3 Neyman–Pearson lemma and uniformly most powerful tests

For a simple null versus simple alternative, $H_0 : \theta = \theta_0$ and $H_1 : \theta = \theta_1$, the Neyman–Pearson lemma says the most powerful level-α test rejects where the likelihood ratio

$$R(X) = \frac{L(\theta_1; X)}{L(\theta_0; X)}$$

is sufficiently large, with randomization if needed at a likelihood-ratio tie. If the same rejection region works for all parameter values in a composite alternative and maintains level under the full null, it is uniformly most powerful (UMP).

A common route to one-sided UMP tests is the monotone likelihood-ratio property: when $R(X)$ is monotone in a statistic $T(X)$ for every $\theta_1 > \theta_0$, the same upper- or lower-tail threshold applies throughout the alternative. A boundary null distribution often determines the least-favorable size, but that boundary claim must be checked rather than assumed.

8.4 Generalized likelihood-ratio testing and Wilks' limit

For composite hypotheses, let $\widehat{\theta}$ maximize likelihood over Θ and $\widetilde{\theta}$ maximize over Θ_0. The generalized likelihood ratio is

$$\Lambda = \frac{L(\widetilde{\theta})}{L(\widehat{\theta})} \leq 1, \qquad D = -2 \log \Lambda \geq 0.$$

Reject for small Λ or large deviance D. If a regular model has r independent smooth restrictions, $D \Rightarrow \chi^2_r$ under H_0 (Wilks' theorem). This is not universally exact and may fail at boundary parameters or when support depends on the unknown parameter. When an exact null distribution is available, it is preferable to a large-sample approximation.

8.5 Why null restrictions create familiar t-tests

In a normal sample with unknown variance, fixing $\mu = \mu_0$ increases the residual sum of squares by $n(\bar{X} - \mu_0)^2$. The ratio of restricted to unrestricted residual variances becomes $1 + T^2/(n-1)$, where T is the one-sample Student statistic. The likelihood ratio is a monotone decreasing function of T^2, so a two-sided normal-mean LRT is exactly the conventional two-sided t-test. With two normal samples and common unknown variance, the analogous decomposition produces the pooled two-sample t-statistic and $n+m-2$ degrees of freedom.

8.6 Nonregular tests and parameter-dependent support

In Uniform$(0, \theta)$ and shifted exponential models, the likelihood is zero whenever data violate parameter-dependent support. A test that rejects where the null likelihood is zero may have zero type-I error cost; the most powerful test may therefore use a nonstandard rejection region and randomization on a likelihood-ratio tie. For shifted exponential location, if the scale is known, one can construct a two-sided UMP test by exploiting such ties. If the scale is unknown, a ratio of the sample minimum to the remaining spacings may be pivotal under H_0, but exact null calibration alone does not prove a UMP claim.

Key result

Testing workflow: (1) state H_0, H_1 and parameter constraints; (2) derive likelihood and sufficient statistic; (3) decide NP or generalized LR; (4) determine null distribution and size; (5) state critical region; (6) derive power or p-value; (7) distinguish exact and asymptotic tests.

P2–1 Exponential rate, exact power and NP test

Prompt. $X_i \overset{\text{iid}}{\sim} \text{Exp}(\text{rate } \theta)$. (a) For $H_0 : \theta \le 1$ vs $H_1 : \theta > 1$, reject if $X_1 \le c$: find power and size. (b) With $H_0 : \theta = 1$, $H_1 : \theta = 2$ and $c = 1$, find Type I/II errors. (c) Prove the size-α NP test for any $\theta_1 > 1$ rejects for small $\sum X_i$; (d) give its power function.
Answers. $\gamma_c(\theta) = 1 - e^{-c\theta}$, size $\alpha = 1 - e^{-c}$ for $c \ge 0$. For part (b), $\alpha = 1 - e^{-1}$, $\beta = e^{-2}$. For parts (c)–(d), write $T = \sum_i X_i$; the LR is $\theta_1^n e^{-(\theta_1 - 1)T}$, decreasing in T. If $c_\alpha = F^{-1}_{\text{Gamma}(n,\text{rate } 1)}(\alpha)$ (equivalently the α-quantile of Gamma$(n, \text{scale } 1)$),

$$\text{reject when } T \le c_\alpha, \qquad \gamma(\theta) = \mathbb{P}_\theta(T \le c_\alpha) = F_{\text{Gamma}(n,\text{rate } \theta)}(c_\alpha).$$

This region is MP for every fixed $\theta_1 > 1$ and therefore UMP for the one-sided alternative.

Part (a): power and least favorable null. For one observation $X_1 \sim \text{Exp}(\theta)$ and the rejection set $\{X_1 \le c\}$,

$$\gamma_c(\theta) = \int_0^c \theta e^{-\theta x} dx = 1 - e^{-\theta c}.$$

For $c > 0$, $\gamma_c'(\theta) = c e^{-\theta c} > 0$. Therefore $\sup_{\theta \le 1} \gamma_c(\theta) = \gamma_c(1)$, which proves the size statement for the composite null.
Part (b): two error probabilities. Under $\theta = 1$, the rejection probability is $1 - e^{-1}$. Under $\theta = 2$, not rejecting means $X_1 > 1$, so $\beta = \mathbb{P}_2(X_1 > 1) = e^{-2}$.
Parts (c)–(d): likelihood ratio and exact calibration. For the whole sample,

$$L(\theta) = \theta^n \exp\left(-\theta \sum_i x_i\right), \qquad \frac{L(\theta_1)}{L(1)} = \theta_1^n \exp[-(\theta_1 - 1)T], \quad T = \sum_i X_i.$$

For any $\theta_1 > 1$ this decreases in T, hence the NP rejection set has the form $\{T \le c_\alpha\}$. Under the boundary null $\theta = 1$, $T \sim \Gamma(n, \text{rate } 1)$. Set $\mathbb{P}_1(T \le c_\alpha) = \alpha$, or $c_\alpha = F^{-1}_{\Gamma(n,1)}(\alpha)$. Under a general rate θ, $\theta T \sim \Gamma(n, 1)$, so

$$\gamma(\theta) = F_{\Gamma(n,1)}(\theta c_\alpha),$$

which is increasing in θ. The rejection threshold is identical for each simple alternative $\theta_1 > 1$, proving the one-sided UMP property.

P2–3 (testing part) Uniform upper-tail UMP test

Prompt. For $X_i \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta)$ test $H_0 : \theta \le \theta_0$ vs $H_1 : \theta > \theta_0$ at size α.
Answers. With $M = X_{(n)}$, $\mathbb{P}_\theta(M \le m) = (m/\theta)^n$ for $0 < m < \theta$. The least favorable null is $\theta = \theta_0$, so choose $c = \theta_0(1 - \alpha)^{1/n}$ and reject for $M > c$. For $\theta > \theta_0$,

$$\gamma(\theta) = 1 - \left(\frac{c}{\theta}\right)^n = 1 - (1 - \alpha)\left(\frac{\theta_0}{\theta}\right)^n.$$

The alternative $\theta > \theta_0$ assigns probability to $M > \theta_0$, impossible under the null; the test combines this automatic rejection with an upper-tail region below θ_0. The NP power bound proves uniform optimality. A simple randomized alternative rejects automatically when $M > \theta_0$ and otherwise with probability α.

Power in all parameter regions. Write $c = \theta_0(1 - \alpha)^{1/n}$. Since $M = \max X_i$, for $0 < \theta \le c$ the rejection probability $\mathbb{P}_\theta(M > c) = 0$. For $\theta > c$ it is

$$\gamma(\theta) = 1 - \mathbb{P}_\theta(M \le c) = 1 - \left(\frac{c}{\theta}\right)^n.$$

In particular, for $\theta \le \theta_0$, the maximum rejection probability is attained at θ_0 and equals α. For $\theta > \theta_0$, substitute $c = \theta_0(1 - \alpha)^{1/n}$ to obtain $\gamma(\theta) = 1 - (1 - \alpha)(\theta_0/\theta)^n$.

Why the NP lemma allows a tail cutoff. For a fixed $\theta_1 > \theta_0$, the ratio $L(\theta_1)/L(\theta_0)$ is infinite on the set $\theta_0 < M \le \theta_1$, which is impossible under the null, and constant on the common support $M \le \theta_0$. Reject all points outside the null support at no size cost; within it any region of null probability α attains the same power against θ_1. The upper-tail region $M > c$ uses this freedom and simultaneously controls every $\theta \le \theta_0$. Thus its UMP claim is justified despite the likelihood-ratio tie.

P2–4 Two independent normal samples: pooled t from the LRT

Prompt. $X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\mu_1, \sigma^2)$, $Y_1, \dots, Y_m \overset{\text{iid}}{\sim} N(\mu_2, \sigma^2)$, independent; common variance unknown. Derive the LRT of $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$ via unrestricted and restricted MLEs.

Sketch and result. Let $SS_W = \sum_i (X_i - \bar{X})^2 + \sum_j (Y_j - \bar{Y})^2$ and $\bar{Z} = (n\bar{X} + m\bar{Y})/(n+m)$. Under H_1, $\widehat{\mu}_1 = \bar{X}$, $\widehat{\mu}_2 = \bar{Y}$, $\widehat{\sigma}^2 = SS_W/(n+m)$. Under H_0, $\widetilde{\mu}_1 = \widetilde{\mu}_2 = \bar{Z}$ and

$$\widetilde{\sigma}^2 = \frac{SS_W + \frac{nm}{n+m}(\bar{X} - \bar{Y})^2}{n+m}.$$

Write $s_p^2 = SS_W/(n+m-2)$ and $T = (\bar{X} - \bar{Y})/(s_p\sqrt{1/n + 1/m})$. The LRT simplifies to

$$\Lambda = \left(1 + \frac{T^2}{n+m-2}\right)^{-(n+m)/2}.$$

Hence reject when $|T| > t_{n+m-2,\,1-\alpha/2}$. Under H_0, this is the exact pooled two-sample t test.

Detailed unrestricted and restricted maximization. Let $N = n + m$ and define

$$Q(\mu_1, \mu_2) = \sum_{i=1}^{n} (X_i - \mu_1)^2 + \sum_{j=1}^{m} (Y_j - \mu_2)^2.$$

Apart from a constant, the log-likelihood is $-N \log \sigma - Q/(2\sigma^2)$. For fixed (μ_1, μ_2) the maximizing variance is $\sigma^2 = Q/N$. Unrestricted minimization of Q gives the separate means $\widehat{\mu}_1 = \bar{X}$ and $\widehat{\mu}_2 = \bar{Y}$, so $Q_1 = SS_W$. Under $H_0 : \mu_1 = \mu_2 = \mu$, differentiation of

$$Q(\mu, \mu) = \sum_i (X_i - \mu)^2 + \sum_j (Y_j - \mu)^2$$

gives the pooled mean $\widetilde{\mu} = (n\bar{X} + m\bar{Y})/N$. The identity

$$Q_0 = SS_W + \frac{nm}{N}(\bar{X} - \bar{Y})^2$$

comes from expanding around the two sample means; all cross terms cancel because the within-group deviations sum to zero.

Likelihood ratio and Student reduction. At their respective variance MLEs, the exponential factors are both $e^{-N/2}$, so

$$\Lambda = (Q_1/Q_0)^{N/2} = \left(1 + \frac{nm}{N}\frac{(\bar{X} - \bar{Y})^2}{SS_W}\right)^{-N/2}.$$

With pooled sample variance $S_p^2 = SS_W/(N-2)$ and $T = (\bar{X} - \bar{Y})/(S_p\sqrt{1/n + 1/m})$, algebra gives

$$\frac{nm}{N}\frac{(\bar{X} - \bar{Y})^2}{SS_W} = \frac{T^2}{N-2}, \qquad \Lambda = \left(1 + \frac{T^2}{N-2}\right)^{-N/2}.$$

Under independent normal sampling with a common variance and H_0, $T \sim t_{N-2}$. The likelihood ratio decreases with T^2, so the critical region is the usual two-sided pooled t rule $|T| > t_{N-2,1-\alpha/2}$.

P2–6 Shifted exponential with unknown scale and $X_{(1)}$

Prompt. For $f(x \mid a, \theta) = \theta^{-1} e^{-(x-a)/\theta} \mathbf{1}\{x \geq a\}$, $\theta > 0$: (a) use exponential spacings to find the law of $W = \sum_i (X_i - X_{(1)})$ and independence from $X_{(1)}$; (b) find MLE of (a, θ); (c)–(e) discuss UMP testing of $a = a_0$ against two-sided alternatives and the power of two-tail tests based on $Y = (X_{(1)} - a_0)/W$.
Results. Independent order-statistic spacings give $2W/\theta \sim \chi^2_{2(n-1)}$ and $X_{(1)} \perp W$. Maximizing over $a \leq X_{(1)}$ yields $(\widehat{a}, \widehat{\theta}) = (X_{(1)}, \bar{X} - X_{(1)})$ (for $n \geq 2$ and nonidentical observations). Under $a = a_0$,

$$U = X_{(1)} - a_0 \sim \text{Exp(rate } n/\theta), \quad \mathbb{P}(Y \leq y) = 1 - (1 + ny)^{-(n-1)}, \ y \geq 0.$$

For $a_1 < a_0$ and a level-α rejection $Y \leq c_1$ or $Y \geq c_2$ with $0 \leq c_1 < c_2$, the power calculation in the computation gives

$$\gamma(a_1) = 1 - (1 - \alpha) e^{-n(a_0 - a_1)/\theta}.$$

Testing conclusion, with an important distinction. *Known θ:* a size-α UMP two-sided test *does exist*: reject if $X_{(1)} < a_0$ or $X_{(1)} > a_0 - (\theta/n)\log\alpha$. Under $a_1 < a_0$, the first region costs no Type I error and the likelihood ratio is constant on the null support; under $a_1 > a_0$, the upper tail is NP-most powerful. *Unknown θ:* Y is pivotal under H_0 and calibrates level-α tests, but the argument does not establish UMP for a general two-sided alternative.

Part (a): exponential spacings. Let $X_{(1)} < \cdots < X_{(n)}$ be ordered observations. Set $D_1 = X_{(1)} - a$ and $D_j = X_{(j)} - X_{(j-1)}$ for $j \geq 2$. The memoryless property makes these independent with $D_j \sim$ Exp(rate $(n - j + 1)/\theta$). Since

$$W = \sum_{i=1}^{n} (X_i - X_{(1)}) = \sum_{j=2}^{n} (n - j + 1) D_j,$$

each summand $(n - j + 1)D_j \sim$ Exp(rate $1/\theta$). Thus $W/\theta \sim \Gamma(n - 1, \text{rate } 1)$, $2W/\theta \sim \chi^2_{2(n-1)}$. Also $X_{(1)} = a + D_1$, which depends only on D_1, so it is independent of W.
Part (b): MLE. The joint likelihood is

$$L(a, \theta) = \theta^{-n} \exp\left[-\frac{\sum_i x_i - na}{\theta}\right] \mathbf{1}\{a \leq x_{(1)}\}.$$

For fixed θ, likelihood increases with a on its admissible region, giving $\widehat{a} = x_{(1)}$. With this choice, $\ell(\theta) = -n\log\theta - W/\theta$. Differentiating gives $\widehat{\theta} = W/n = \bar{X} - x_{(1)}$, provided $W > 0$.
Part (c): known-scale test. If $H_0 : a = a_0$, then $X_{(1)} - a_0 \sim$ Exp(rate n/θ). The event $X_{(1)} < a_0$ is impossible under the null and therefore can always be included in rejection. On the null support, choose $c = a_0 - (\theta/n)\log\alpha$ to make $\mathbb{P}_{a_0}(X_{(1)} > c) = \alpha$. Reject when $X_{(1)} < a_0$ or $X_{(1)} > c$. The likelihood-ratio argument gives the UMP property for the two-sided location alternative when θ is fixed and known.
Part (d): unknown-scale pivot. Under $a = a_0$, $A = n(X_{(1)} - a_0)/\theta \sim \text{Exp}(1)$, independent of $B = W/\theta \sim \Gamma(n - 1, \text{rate } 1)$. For $Y = (X_{(1)} - a_0)/W = A/(nB)$ and $y \geq 0$,

$$\mathbb{P}(Y > y) = \mathbb{E}[e^{-nyB}] = (1 + ny)^{-(n-1)}.$$

Therefore $F_Y(y) = 1 - (1 + ny)^{-(n-1)}$, a parameter-free null CDF. A two-tailed level-α test can be calibrated from quantiles of this distribution, assigning tail probabilities α_1 and α_2 with $\alpha_1 + \alpha_2 = \alpha$. But without additional optimality work, a pivotal test is not automatically UMP against the two-sided composite alternative.
Part (e): parameter and nuisance distinction. The existence of a UMP test depends on which parameters are fixed and which are unknown. A known θ yields a simple support-based construction; an unknown θ introduces nuisance estimation and changes the comparison class. Always state the model assumptions before declaring that a UMP test does or does not exist.

Likelihood-ratio tie: The naive argument asserts that no two-sided UMP test exists even when the scale θ is known. That conclusion overlooks a likelihood-ratio tie on the common support: the displayed test is NP-optimal for both directions. The θ-unknown two-sided situation is different and should not be inferred from the known-scale case.

F–4 One-sample normal mean with unknown variance: LRT and t limit

Prompt. For $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, both parameters unknown, test $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$ by LRT. Then prove that the studentized mean is asymptotically $N(0,1)$; contrast normal exact sampling and finite-variance asymptotics.
Derivation. $\widehat{\mu} = \bar{X}$ and $\widehat{\sigma}^2 = n^{-1}\sum(X_i - \bar{X})^2$. Under H_0, $\widetilde{\sigma}^2 = n^{-1}\sum(X_i - \mu_0)^2 = \widehat{\sigma}^2 + (\bar{X} - \mu_0)^2$. With $S^2 = (n-1)^{-1}\sum(X_i - \bar{X})^2$ and $T = \sqrt{n}(\bar{X} - \mu_0)/S$,

$$\Lambda = \left(1 + \frac{T^2}{n-1}\right)^{-n/2}, \qquad -2\log\Lambda = n\log\left(1 + \frac{T^2}{n-1}\right).$$

Under normal sampling and H_0, $T \sim t_{n-1}$ exactly. Under general i.i.d. finite positive variance, CLT gives $\sqrt{n}(\bar{X} - \mu)/\sigma \xrightarrow{d} N(0,1)$; LLN gives $S/\sigma \xrightarrow{p} 1$; hence $T \xrightarrow{d} N(0,1)$ by Slutsky. Under standard regularity, $-2\log\Lambda \xrightarrow{d} \chi^2_1$.

Complete LRT algebra. Let $RSS_1 = \sum_i (X_i - \bar{X})^2$ be the unrestricted residual sum of squares. Under the null mean μ_0, write

$$RSS_0 = \sum_i (X_i - \mu_0)^2 = RSS_1 + n(\bar{X} - \mu_0)^2.$$

The maximum likelihood variances are $\widehat{\sigma}_1^2 = RSS_1/n$ and $\widehat{\sigma}_0^2 = RSS_0/n$, so

$$\Lambda = (RSS_1/RSS_0)^{n/2} = \left(1 + \frac{n(\bar{X} - \mu_0)^2}{RSS_1}\right)^{-n/2}.$$

With $S^2 = RSS_1/(n-1)$ and $T = \sqrt{n}(\bar{X} - \mu_0)/S$, the ratio is $[1 + T^2/(n-1)]^{-n/2}$ and $D = n\log[1 + T^2/(n-1)]$. The rejection rule $D > d$ is equivalent to $|T| > t$ for an appropriate threshold.
Proof A: CLT plus SLLN. For general iid data with finite positive variance, $Z_n = \sqrt{n}(\bar{X} - \mu)/\sigma \Rightarrow N(0,1)$. The SLLN applied to X_i and X_i^2 gives $S^2 \to \sigma^2$ almost surely. Therefore $T_n = Z_n(\sigma/S) \Rightarrow N(0,1)$.
Proof B: characteristic functions. Let $V_n = S/\sigma \to 1$ in probability. Because the exponential map is bounded, for fixed t the characteristic functions of Z_n/V_n approach that of Z by Slutsky and bounded convergence arguments. The limit is $e^{-t^2/2}$, the standard normal characteristic function.
Proof C: exact Gaussian decomposition. Under normal sampling, $Z = \sqrt{n}(\bar{X} - \mu)/\sigma \sim N(0,1)$ and $Q = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, independent. Therefore $T = Z/\sqrt{Q/(n-1)} \sim t_{n-1}$ exactly. As $n \to \infty$, $Q/(n-1) \to 1$ in probability, and the exact Student law converges to standard normal. The finite-sample exact conclusion needs normality; Proof A does not.
Wilks connection. Since $T_n = O_p(1)$ under H_0, let $u = T_n^2/(n-1) = O_p(1/n)$. Then $D = n\log(1+u) = nu + O_p(nu^2) = T_n^2 + o_p(1)$, yielding $D \Rightarrow Z^2 \sim \chi^2_1$.

F–6 Shifted exponential with known scale: location LRT

Prompt. For $f(x \mid \theta) = \frac{1}{2}e^{-(x-\theta)/2}\mathbf{1}\{x \geq \theta\}$, find a level-$\alpha$ likelihood-ratio test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.
Answer. The unrestricted location MLE is $X_{(1)}$. For data possible under H_0, the full/null likelihood ratio is $e^{n(X_{(1)} - \theta_0)/2}$. Since $X_{(1)} - \theta_0 \sim \text{Exp}(\text{rate } n/2)$ under H_0,

$$\mathbb{P}_{\theta_0}(X_{(1)} > c) = e^{-n(c-\theta_0)/2} = \alpha \implies \boxed{c = \theta_0 - \tfrac{2}{n}\log\alpha}.$$

Reject for $X_{(1)} > c$; also reject automatically if $X_{(1)} < \theta_0$, a zero-probability event under H_0 but possible under alternatives. Including the support-violating region makes the operational test complete.

Likelihood and null support. For known scale 2, the joint likelihood is

$$L(\theta) = 2^{-n} \exp\left[-\frac{\sum_i x_i - n\theta}{2}\right] \mathbf{1}\{\theta \le x_{(1)}\}.$$

It increases with θ on the allowed interval, so the unrestricted MLE is $\widehat{\theta} = x_{(1)}$. On the support $x_{(1)} \ge \theta_0$ of the null, the ratio $L(\widehat{\theta})/L(\theta_0) = \exp[n(x_{(1)} - \theta_0)/2]$ increases with the minimum. Therefore the LRT rejects for large $x_{(1)}$; if $x_{(1)} < \theta_0$ it also rejects because the null assigns zero probability to such data.

Exact cutoff. Under H_0, the survival function of $X_{(1)}$ is $\mathbb{P}_{\theta_0}(X_{(1)} > x) = \exp[-n(x - \theta_0)/2]$, $x \ge \theta_0$. Setting this equal to α gives $c = \theta_0 - (2/n)\log\alpha$. The test is exact level α, with rejection set $\{x_{(1)} < \theta_0\} \cup \{x_{(1)} > c\}$.

F–7 Two-sided LRT for exponential rate

Prompt. If $X_i \overset{\text{iid}}{\sim}$ Exp(rate λ), derive an LRT for $H_0 : \lambda = \lambda_0$ vs $H_1 : \lambda \ne \lambda_0$; compare exact critical values and the large-sample test.

Answer. For $T = \sum_i X_i$, $\widehat{\lambda} = n/T$ and

$$\Lambda = \left(\frac{\lambda_0 T}{n}\right)^n e^{n - \lambda_0 T}, \qquad D = -2\log\Lambda = 2\left[\lambda_0 T - n - n\log\frac{\lambda_0 T}{n}\right].$$

Under H_0, $T \sim$ Gamma$(n,$ rate $\lambda_0)$. Reject for $D > k_\alpha$, equivalently $T < c_1$ or $T > c_2$, where $D(c_1) = D(c_2) = k_\alpha$ and $\mathbb{P}_{\lambda_0}(T < c_1) + \mathbb{P}_{\lambda_0}(T > c_2) = \alpha$. In large samples use $D \dot{\sim} \chi^2_1$.

Distinguish the two exact tests: the computation also gives an *equal-tailed gamma test*, with $c_1 = F_T^{-1}(\alpha/2)$ and $c_2 = F_T^{-1}(1 - \alpha/2)$; this controls size exactly but generally does *not* use a single likelihood-ratio cutoff. Both constructions are listed because both procedures are useful and mathematically distinct. For its numerical case $n = 3$, $\lambda_0 = 2$, $\alpha = .05$, the exact equal-tail gamma cutoffs are approximately $c_1 = 0.309$ and $c_2 = 3.612$.

Exact likelihood-ratio shape. For a rate parameter λ,

$$\ell(\lambda) = n\log\lambda - \lambda T, \qquad \widehat{\lambda} = n/T.$$

The ratio of null and unrestricted maxima equals

$$\Lambda(T) = \frac{\lambda_0^n e^{-\lambda_0 T}}{(n/T)^n e^{-n}} = \left(\frac{\lambda_0 T}{n}\right)^n e^{n - \lambda_0 T}.$$

Let $u = \lambda_0 T/n$. The deviance is $D = 2n(u - 1 - \log u)$; the function $u - 1 - \log u$ has minimum zero at $u = 1$ and grows on either side. Therefore D rejects for very small or very large T.

Exact LRT calibration. Under H_0, T has a Gamma$(n,$ rate $\lambda_0)$ law. Solve for $a < b$ and k satisfying

$$D(a) = D(b) = k, \qquad F_{T,0}(a) + 1 - F_{T,0}(b) = \alpha.$$

The two cutoff points generally do not allocate probability $\alpha/2$ to each tail. An easier exact *equal-tail* alternative instead chooses $a = F_{T,0}^{-1}(\alpha/2)$ and $b = F_{T,0}^{-1}(1 - \alpha/2)$; this has exactly the prescribed level but is not in general the LRT because the likelihood ratios at its two endpoints differ.

Asymptotic approximation. Expand $u - 1 - \log u = \frac{1}{2}(u - 1)^2 + o((u - 1)^2)$ near $u = 1$. Since $\sqrt{n}(u - 1) \Rightarrow N(0, 1)$ under H_0, $D \Rightarrow \chi^2_1$, yielding the large-sample chi-square cutoff. The exact gamma construction is preferable at small n.

Numerical check: Approximate cutoffs 0.32 and 3.87 are not equal-tailed 5% Gamma$(3, \text{rate } 2)$ quantiles; direct Gamma$(3, \text{rate } 2)$ quantiles give 0.309 and 3.612 for an equal-tail 5% test.

9 Linear Regression, Score Vector and Multiplier Geometry

9.1 Linear regression under second-moment assumptions

Consider $\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with fixed $n \times p$ full-rank X, $\mathbb{E}\boldsymbol{\varepsilon} = 0$ and $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 I_n$. Least squares minimizes $\|\boldsymbol{y} - X\boldsymbol{\beta}\|_2^2$, giving the normal equations $X^\top X\widehat{\boldsymbol{\beta}} = X^\top \boldsymbol{y}$. Gaussianity is *not required* for these first- and second-moment results.

9.2 Least squares geometry, normal equations and orthogonal projections

Let $y = X\beta + \varepsilon$ with full-column-rank design matrix $X \in \mathbb{R}^{n\times p}$ ($n \geq p$). Least squares minimizes

$$Q(b) = (y - Xb)^\top (y - Xb) = y^\top y - 2b^\top X^\top y + b^\top X^\top Xb.$$

Differentiating yields $\nabla_b Q = -2X^\top y + 2X^\top Xb$. Since $X^\top X$ is positive definite, the unique minimizer is

$$\widehat{\beta} = (X^\top X)^{-1} X^\top y.$$

The fitted response is $\widehat{y} = Hy$ where $H = X(X^\top X)^{-1}X^\top$ is the orthogonal projection onto the column space of X. It obeys $H^\top = H$, $H^2 = H$ and $X^\top(y - \widehat{y}) = 0$. The residual $y - \widehat{y}$ is orthogonal to every design column.

9.3 Gauss–Markov theorem and efficient linear unbiased estimation

Assume $\mathbb{E}\varepsilon = 0$ and $\mathrm{Cov}(\varepsilon) = \sigma^2 I$, with no need for normally distributed errors. Then $\mathbb{E}\widehat{\beta} = \beta$ and $\mathrm{Cov}(\widehat{\beta}) = \sigma^2(X^\top X)^{-1}$. Any linear estimator Ay is unbiased for every β if and only if $AX = I_p$. Write

$$A = (X^\top X)^{-1}X^\top + B, \qquad BX = 0.$$

The covariance difference becomes $\sigma^2 BB^\top \succeq 0$, since $v^\top BB^\top v = \|B^\top v\|^2 \geq 0$. This proves that least squares is BLUE: *best linear unbiased* in covariance order. The theorem does not assert that least squares beats every nonlinear or biased estimator.

9.4 Score gradient and directional likelihood change

For $\theta \in \mathbb{R}^k$, the score is $s(\theta) = \nabla_\theta \ell(\theta)$. The first-order likelihood change from a small move h is

$$\ell(\theta + h) - \ell(\theta) \approx s(\theta)^\top h = \|s(\theta)\|\,\|h\| \cos\gamma.$$

For fixed $\|h\|$, this quantity is maximal in the direction of $s(\theta)$. The score is therefore the direction of steepest local ascent. At an unconstrained smooth interior MLE, $s(\widehat{\theta}) = 0$.

9.5 Lagrange multiplier as the shadow price of a constraint

Suppose we maximize $\ell(\theta)$ subject to $g(\theta) = \varepsilon$. Let $G(\theta) = Dg(\theta)$ and use the Lagrangian $\mathcal{L}(\theta, \lambda) = \ell(\theta) - \lambda^\top[g(\theta) - \varepsilon]$. The first-order conditions are

$$\nabla\ell(\theta^*) = G(\theta^*)^\top \lambda^*, \qquad g(\theta^*) = \varepsilon.$$

If $V(\varepsilon) = \ell(\theta^*(\varepsilon))$ is the optimized value, the envelope calculation gives

$$\nabla_\varepsilon V(\varepsilon) = \lambda^*.$$

Thus one multiplier component measures the first-order change in the maximized log likelihood when the corresponding constraint is relaxed. A sign change in the Lagrangian definition reverses the multiplier sign; the mathematical meaning of the shadow price remains the same.

The vector identity $u^\top v = \|u\|\|v\| \cos\gamma$ follows by comparing $\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2u^\top v$ to the law of cosines. This connects the likelihood gradient picture with directional derivatives.

9.6 Three classical likelihood tests: Wald, likelihood ratio and score

Suppose $H_0 : g(\theta) = 0$ imposes r smooth constraints. A *Wald test* measures the estimated distance between an unrestricted MLE and the null; a *likelihood-ratio test* compares optimized likelihoods in unrestricted and restricted models; a *score or Lagrange-multiplier test* evaluates how strongly the restricted fit wants to move away from the null using its score and Fisher information. Under regular conditions, all three tests have asymptotic χ^2_r null laws and agree to first order, but may differ for small samples and nonregular models.

P1–6 Least squares and Gauss–Markov theorem

Prompt. (a) Find the least-squares coefficient estimate, its expectation and covariance; (b) give the necessary and sufficient condition for $A\boldsymbol{y}$ to be unbiased for $\boldsymbol{\beta}$ for all $\boldsymbol{\beta}$; (c) prove its covariance exceeds least squares in the positive-semidefinite order.

Answers.

$$\widehat{\boldsymbol{\beta}}_{LS} = (X^\top X)^{-1} X^\top \boldsymbol{y}, \quad \mathbb{E}\widehat{\boldsymbol{\beta}}_{LS} = \boldsymbol{\beta}, \quad \mathrm{Cov}(\widehat{\boldsymbol{\beta}}_{LS}) = \sigma^2 (X^\top X)^{-1}.$$

An arbitrary linear estimator $A\boldsymbol{y}$ is unbiased exactly when $AX = I_p$. Set $B = A - (X^\top X)^{-1} X^\top$. Then $BX = 0$ and $AA^\top = (X^\top X)^{-1} + BB^\top$. Thus

$$\mathrm{Cov}(A\boldsymbol{y}) - \mathrm{Cov}(\widehat{\boldsymbol{\beta}}_{LS}) = \sigma^2 BB^\top \succeq 0.$$

This is the best-linear-unbiased-estimator (BLUE) property.

Part (a): minimization and sampling covariance. Expand $Q(b) = \|y - Xb\|^2$, differentiate to obtain $X^\top Xb = X^\top y$, and solve using invertibility of $X^\top X$. Substituting $y = X\beta + \varepsilon$ yields

$$\widehat{\beta} = \beta + (X^\top X)^{-1} X^\top \varepsilon.$$

The error has mean zero. Its covariance is

$$\mathrm{Cov}(\widehat{\beta}) = (X^\top X)^{-1} X^\top (\sigma^2 I) X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}.$$

Part (b): condition for any linear unbiased estimator. If $\delta = Ay$ for a fixed $p \times n$ matrix A, then $\mathbb{E}_\beta \delta = AX\beta$. It equals β for every vector β if and only if $AX = I_p$.

Part (c): comparison for every direction. Set $A_0 = (X^\top X)^{-1} X^\top$, and write $A = A_0 + B$. Since $AX = I = A_0 X$, we get $BX = 0$. Cross terms vanish in

$$AA^\top = (A_0 + B)(A_0 + B)^\top = A_0 A_0^\top + BB^\top = (X^\top X)^{-1} + BB^\top.$$

Therefore for every $v \in \mathbb{R}^p$,

$$v^\top \{\mathrm{Cov}(Ay) - \mathrm{Cov}(\widehat{\beta})\} v = \sigma^2 \|B^\top v\|^2 \geq 0.$$

This proves the matrix positive-semidefinite ordering and the BLUE theorem without requiring Gaussian errors.

9.7 Score as steepest ascent and multiplier as shadow price

For parameter vector $\boldsymbol{\theta} \in \mathbb{R}^k$, the *score vector* is $s(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta})$. A local perturbation obeys $\ell(\boldsymbol{\theta} + \Delta\boldsymbol{\theta}) - \ell(\boldsymbol{\theta}) \approx s(\boldsymbol{\theta})^\top \Delta\boldsymbol{\theta}$. Because $s^\top \Delta\theta = \|s\|\,\|\Delta\theta\| \cos\gamma$, the maximum directional increase (for fixed perturbation

length) is along s.

For constrained maximum likelihood $\max \ell(\boldsymbol{\theta})$ subject to $g(\boldsymbol{\theta}) = \boldsymbol{\varepsilon}$, the first-order Lagrange condition is $s(\boldsymbol{\theta}^*) = G(\boldsymbol{\theta}^*)^\top \boldsymbol{\lambda}^*$, where G is the Jacobian of g. If $V(\boldsymbol{\varepsilon})$ is the constrained optimum, differentiation gives $\nabla_{\boldsymbol{\varepsilon}} V = \boldsymbol{\lambda}^*$ (with the appendix's chosen sign convention). Thus the multiplier measures the first-order *shadow price* of relaxing the constraint. The appendix derives $u^\top v = \|u\|\|v\| \cos\gamma$ from the law of cosines.

Key result

Connections to statistical tests: a likelihood ratio compares unrestricted and restricted maxima; a score / Lagrange-multiplier test asks how strongly the *restricted* optimum wants to move off the null constraint. The Week 12 PDF gives the geometric interpretation, not a standalone numeric practice exercise.

Formula and Method Reference

Which tool matches which problem?

Clue	First move
Event / partition / conditioning	Write A, B; use $\mathbb{P}(A \mid B)$, total probability or Bayes.
Nonlinear measurement $Y = g(X)$	Solve inverse branches; apply Jacobian or CDF transformation.
Counts of successes or events	Binomial / multinomial / Poisson; condition on totals to eliminate intensity.
Waiting to the kth event	Negative binomial or gamma; exploit $\{N(t) \geq k\} = \{W_k \leq t\}$.
Large-n limit of sums	LLN for denominator, CLT for centered sum, Slutsky for ratio.
Find estimator	Form likelihood including parameter-dependent support; differentiate or use order statistics.
Best unbiased estimator	Seek sufficient/complete T; condition U on T (Rao–Blackwell).
Find a confidence interval	Identify parameter-free pivot and invert known quantiles.
Test H_0 vs H_1	Define size/power; NP for simple, LR for composite; determine null reference law.
Constrained optimization	Compare restricted/full likelihoods; relate score to normal direction of constraint.
Linear least squares	Normal equations; use $AX = I$ and covariance PSD comparison.